

ETH ZÜRICH

BACHELOR THESIS

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# Beyond Quadratic Reciprocity

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## Abstract

Quadratic reciprocity was first conjectured by Euler and Dirichlet, and then proved by Gauss in the *Disquisitiones Arithmeticae*. Gauss referred to it as *theorema aureum*, the “golden theorem” and there are almost 200 different proofs of it. Later, Euler and Gauss also found a cubic and biquadratic reciprocity law. More than a hundred years later, Emil Artin discovered a reciprocity law in class field theory that encompasses all known reciprocity laws.

This bachelor thesis follows the chronological journey of the development of reciprocity laws for power residue symbols. It aims to explore some fundamental concepts of class field theory and illustrate how Artin’s generalized reciprocity law implies the quadratic and cubic reciprocity laws.

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# 1 Quadratic Reciprocity

**Definition 1.1.** Let  $p \geq 3$  be a prime number, and let  $a \in \mathbb{Z}$  be coprime to  $p$ . Then  $a$  is a quadratic residue mod  $p$  if there exists  $x \in \mathbb{Z}$  such that  $x^2 \equiv a \pmod{p}$ .

**Example 1.2.** (i) 2 is a quadratic residue mod 7 since  $3^2 = 9 \equiv 2 \pmod{7}$ .

(ii) 3 not a quadratic residue mod 7.

**Remark 1.3.** In fact, precisely  $\frac{p-1}{2}$  elements are quadratic residues mod  $p$  if  $p$  is an odd prime.

**Lemma 1.4.** Let  $p \geq 3$  be a prime number, and let  $a \in \mathbb{Z}$  be coprime to  $p$ . Then the following are equivalent:

- (i)  $a$  is a quadratic residue mod  $p$ .
- (ii)  $a^{\frac{p-1}{2}} \equiv 1 \pmod{p}$ .

*Proof.* If  $a$  is a quadratic residue mod  $p$ , then let  $x$  be such that  $x^2 \equiv a \pmod{p}$ . By Fermat's little theorem,

$$a^{\frac{p-1}{2}} \equiv x^{p-1} \equiv 1 \pmod{p}.$$

Conversely, assume  $a^{\frac{p-1}{2}} \equiv 1 \pmod{p}$ . Choose a generator  $g$  of  $(\mathbb{Z}/p\mathbb{Z})^*$ , and let  $r \in \mathbb{N}$  such that  $a \equiv g^r \pmod{p}$ . Now  $g^{\frac{r(p-1)}{2}} \equiv 1 \pmod{p}$ , so  $p-1 \mid \frac{r(p-1)}{2}$ . This implies  $r$  is even, and  $x = g^{r/2}$  satisfies  $x^2 \equiv a \pmod{p}$ .  $\square$

**Definition 1.5.** Let  $p$  be an odd prime, and let  $a \in \mathbb{Z}$ . Define the quadratic Legendre symbol to be

$$\left(\frac{a}{p}\right) = \begin{cases} 1 & \text{if } p \nmid a \text{ and } a \text{ is a quadratic residue mod } p \\ -1 & \text{if } p \nmid a \text{ and } a \text{ is not a quadratic residue mod } p \\ 0 & \text{if } p \mid a. \end{cases}$$

**Lemma 1.6.** Let  $p \geq 3$  be a prime number, and let  $a, b \in \mathbb{Z}$ . Then:

- (i)  $\left(\frac{a}{p}\right) \equiv a^{\frac{p-1}{2}} \pmod{p}$ ,
- (ii)  $\left(\frac{ab}{p}\right) = \left(\frac{a}{p}\right) \cdot \left(\frac{b}{p}\right)$ ,
- (iii)  $a \equiv b \pmod{p} \implies \left(\frac{a}{p}\right) = \left(\frac{b}{p}\right)$ .

*Proof.* (i) For  $a \nmid p$  this is just Lemma 1.4, using that  $a^{\frac{p-1}{2}} \equiv \pm 1 \pmod{p}$  by Fermat's little theorem. If  $p \mid a$ , then  $a^{\frac{p-1}{2}} \equiv 0 \pmod{p}$ .

(ii) Follows from (i), since

$$(ab)^{\frac{p-1}{2}} \equiv a^{\frac{p-1}{2}} \cdot b^{\frac{p-1}{2}} \pmod{p}.$$

We thus have  $\left(\frac{ab}{p}\right) \equiv \left(\frac{a}{p}\right) \cdot \left(\frac{b}{p}\right) \pmod{p}$ , but both sides are in  $\{0, \pm 1\}$ .

(iii) Also follows from (i). □

**Remark 1.7.** The map  $(\mathbb{Z}/p\mathbb{Z})^* \rightarrow \{\pm 1\}$ ,  $a \mapsto \left(\frac{a}{p}\right)$  is a group homomorphism.

**Proposition 1.8.** Let  $p \geq 3$  be a prime number, then

- (i)  $\left(\frac{-1}{p}\right) = (-1)^{\frac{p-1}{2}}$ ,
- (ii)  $-1$  is a quadratic residue mod  $p$  if and only if  $p \equiv 1 \pmod{4}$ .

*Proof.* The first part follows directly from Lemma 1.6 (iii) since  $(-1)^{\frac{p-1}{2}} \in \{\pm 1\}$ . For part (ii), we note that  $\frac{p-1}{2}$  is even if and only if  $p \equiv 1 \pmod{4}$ . □

We can now state quadratic reciprocity.

**Theorem 1.9** (Quadratic reciprocity). Let  $p, q$  be distinct odd primes. Then

$$\left(\frac{p}{q}\right) = \left(\frac{q}{p}\right) (-1)^{\frac{(p-1)}{2} \cdot \frac{(q-1)}{2}}.$$

We also have a supplementary law for the prime 2.

**Proposition 1.10.** Let  $p$  be an odd prime. Then

$$\left(\frac{2}{p}\right) = (-1)^{\frac{p^2-1}{8}}.$$

We will see proofs of these statements in section 6. But first, let us calculate an explicit example.

**Example 1.11.** Using quadratic reciprocity, we can easily figure out whether 57 is a square mod 107.

$$\begin{aligned} \left(\frac{57}{107}\right) &= \left(\frac{3}{107}\right) \cdot \left(\frac{19}{107}\right) = (-1) \left(\frac{107}{3}\right) \cdot (-1) \left(\frac{107}{19}\right) = \left(\frac{2}{3}\right) \cdot \left(\frac{12}{19}\right) \\ &= (-1) \cdot \left(\frac{2}{19}\right)^2 \cdot \left(\frac{3}{19}\right) = \left(\frac{19}{3}\right) = \left(\frac{1}{3}\right) = 1 \end{aligned}$$

So indeed, 57 is a square mod 107!

## 2 Cubic Reciprocity

It is natural to ask whether more general statements are to be found. Indeed, Euler and Gauss found statements similar to quadratic reciprocity, called the cubic and biquadratic reciprocity law, defined for primes of the rings  $\mathbb{Z}[\omega]$  and  $\mathbb{Z}[i]$ , for  $\omega = e^{\frac{2\pi i}{3}}$ .

In this section we introduce the cubic Legendre symbol and state its properties, mainly following Chapter 1 §4 from Cox's book 'Primes of the form  $x^2 + ny^2$ ' ([Cox89]). Throughout this section, let  $\omega = e^{\frac{2\pi i}{3}}$ .

First note that for  $K = \mathbb{Q}(\omega)$  the ring of integers  $O_K$  is just  $\mathbb{Z}[\omega]$ . Also notice that  $\mathbb{Z}[\omega]$  is a Euclidean ring and hence a PID. In  $\mathbb{Q}(\omega)$  the norm of an element  $\alpha$  is just the usual complex norm squared  $N_{\mathbb{Q}(\omega)/\mathbb{Q}}(\alpha) = \alpha\bar{\alpha}$  since  $\text{Gal}(\mathbb{Q}(\omega)/\mathbb{Q})$  consists of the identity and the map induced by complex conjugation. We will continue by only writing  $N$  for  $N_{\mathbb{Q}(\omega)/\mathbb{Q}}$ .

**Proposition 2.1.** The unit group of  $\mathbb{Z}[\omega]$  is  $\{\pm 1, \pm\omega, \pm\omega^2\}$ .

*Proof.* We have that if  $\alpha \in \mathbb{Z}[\omega]$  satisfies  $N(\alpha) = \alpha\bar{\alpha} = 1$  then clearly  $\alpha$  is on the unit circle and the only solutions in  $\mathbb{Z}[\omega]$  are  $\{\pm 1, \pm\omega, \pm\omega^2\}$ .  $\square$

Recall that two primes  $\pi, \theta$  are called *associate* if there exists a unit  $u$  such that  $\pi = u\theta$ . We now want to classify all primes in  $\mathbb{Z}[\omega]$ .

**Proposition 2.2** ([Cox89], Prop.4.7). Let  $p$  be a prime in  $\mathbb{Z}$ . Then:

- (i) If  $p = 3$ , then  $p = -\omega^2(1 - \omega)^2$  and  $1 - \omega$  is prime in  $\mathbb{Z}[\omega]$ .
- (ii) If  $p \equiv 1 \pmod{3}$ , then  $p = \pi\bar{\pi}$  for some prime  $\pi$  of  $\mathbb{Z}[\omega]$ .
- (iii) If  $p \equiv 2 \pmod{3}$ , then  $p$  is prime in  $\mathbb{Z}[\omega]$ .

Furthermore, every prime  $\pi$  of  $\mathbb{Z}[\omega]$  is associate to one of the primes listed above.

*Proof.* These all follow from some basic algebraic number theory facts. We can check (i) explicitly, using that  $N(1 - \omega) = (1 - \omega)(1 - \bar{\omega}) = 3$ , which means that  $1 - \omega$  is prime. For (ii) and (iii) we use Dedekind's criterion. The minimal polynomial of  $\omega$  is  $X^2 + X + 1$ . It is irreducible modulo  $p \neq 3$  if and only if  $X^3 - 1 = (X^2 + X + 1)(X - 1)$  has no other zeros except 1. The zeros of  $X^3 - 1$  are precisely the elements of order dividing 3 in the multiplicative group  $(\mathbb{Z}/p\mathbb{Z})^* \cong C_{p-1}$ . We see that for  $p \equiv 2 \pmod{3}$  there are no elements of order three in  $(\mathbb{Z}/p\mathbb{Z})^* \cong C_{p-1}$ , so by Dedekind's criterion  $p$  is inert in  $\mathbb{Z}[\omega]$ . For  $p \equiv 1 \pmod{3}$ ,  $X^2 + X + 1$  factors into two distinct linear terms modulo  $p$ . If we write  $p = \pi\theta$  as its unique factorisation into two primes, then  $N(\pi), N(\theta) \mid p^2$  and  $N(\pi), N(\theta) \neq 1$  so  $p = N(\pi) = \pi\bar{\pi}$  and we deduce (ii).

Finally, let  $\pi$  be an arbitrary prime in  $\mathbb{Z}[\omega]$ . Then the norm  $N(\pi) = \pi\bar{\pi}$  factors into primes in  $\mathbb{Z}$  and thus into primes of  $\mathbb{Z}[\omega]$  listed above. By unique factorisation,  $\pi$  is already associate to a prime of the form of one of primes in (i), (ii) or (iii).  $\square$

**Corollary 2.3** ([Cox89], Lemma.4.8). If  $\pi$  is a prime in  $\mathbb{Z}[\omega]$  then  $N(\pi) = p$  or  $p^2$  for some prime  $p$  and

- (i) if  $p = 3$  or  $p \equiv 1 \pmod{3}$ , then  $N(\pi) = p$  and  $\mathbb{Z}/p\mathbb{Z} \cong \mathbb{Z}[\omega]/\pi\mathbb{Z}[\omega]$ .
- (ii) if  $p \equiv 2 \pmod{3}$ , then  $N(\pi) = p^2$  and  $\mathbb{Z}/p\mathbb{Z} \subset \mathbb{Z}[\omega]/\pi\mathbb{Z}[\omega]$  is the unique subfield of order  $p$ .

*Proof.* Follows directly since  $\mathbb{Z}[\omega]/\pi\mathbb{Z}[\omega]$  is a finite field with  $N(\pi)$  elements and  $\mathbb{Z}/p\mathbb{Z} \subset \mathbb{Z}[\omega]/\pi\mathbb{Z}[\omega]$  is a subfield of order  $p$ .  $\square$

Now let  $\pi$  be a prime that is not associate to  $1 - \omega$  and be  $\alpha \in \mathbb{Z}[\omega]$  be not divisible by  $\pi$ . Notice that either  $N(\pi) = p$  with  $p \equiv 1 \pmod{3}$  or  $N(\pi) = p^2$  with  $p \equiv 2 \pmod{3}$ . So in any case  $N(\pi) \equiv 1 \pmod{3}$  and therefore  $(N(\pi) - 1)/3$  is an integer. Fermat's little theorem tells us that

$$\alpha^{N(\pi)-1} \equiv 1 \pmod{\pi}.$$

Thus  $\alpha^{(N(\pi)-1)/3}$  is a cube root of unity modulo  $\pi$ . Moreover, we can deduce that  $1, \omega, \omega^2$  are distinct modulo  $\pi$  if  $\pi$  is not associate to  $1 - \omega$ . In fact, if  $1 - \omega \equiv 0 \pmod{\pi}$  then  $\pi$  is associate to  $1 - \omega$  by unique factorisation, similarly if  $1 - \omega^2 \equiv 0 \pmod{\pi}$  then  $\pi$  is associate to  $1 - \omega^2 = -\omega^2(1 - \omega)$ .

Now we can give the following definition of the cubic Legendre symbol.

**Definition 2.4.** Let  $\pi$  be a prime that is not associate to  $1 - \omega$  and let  $\alpha \in \mathbb{Z}[\omega]$  be not divisible by  $\pi$ . We define the cubic Legendre symbol  $\left(\frac{\alpha}{\pi}\right)_3$  to be the cube root of unity that satisfies

$$\alpha^{\frac{N(\pi)-1}{3}} \equiv \left(\frac{\alpha}{\pi}\right)_3 \pmod{\pi}$$

The cubic Legendre symbol satisfies similar properties as the quadratic Legendre symbol.

**Proposition 2.5.** Let  $\pi$  be a prime not associate to  $1 - \omega$  and let  $\alpha, \beta \in \mathbb{Z}[\omega]$  be not divisible by  $\pi$ . Then

- (i)  $\left(\frac{\alpha\beta}{\pi}\right)_3 = \left(\frac{\alpha}{\pi}\right)_3 \cdot \left(\frac{\beta}{\pi}\right)_3$ ,
- (ii)  $\alpha \equiv \beta \pmod{\pi} \implies \left(\frac{\alpha}{\pi}\right)_3 = \left(\frac{\beta}{\pi}\right)_3$ .

In particular, we have a group homomorphism

$$\left(\frac{\cdot}{\pi}\right) : (\mathbb{Z}[\omega]/\pi\mathbb{Z}[\omega])^* \longrightarrow \{1, \omega, \omega^2\}.$$

*Proof.* Both follow directly from the definition of the cubic Legendre symbol. □

Notice how similar the definition of the cubic Legendre symbol is to the characterisation of the quadratic Legendre symbol in Lemma 1.4. It is natural to wonder whether it has a similar relation to cubes modulo primes as the quadratic Legendre symbol to quadratic residues.

**Proposition 2.6.** Let  $\pi$  be a prime in  $\mathbb{Z}[\omega]$  not associate to  $1 - \omega$  and let  $\alpha \in \mathbb{Z}[\omega]$  be not divisible by  $\pi$ . Then

$$\left(\frac{\alpha}{\pi}\right)_3 = 1 \iff x^3 \equiv \alpha \pmod{\pi} \text{ has a solution in } \mathbb{Z}[\omega]$$

*Proof.* The proof of

$$\left(\frac{\alpha}{\pi}\right)_3 = 1 \implies x^3 \equiv \alpha \pmod{\pi} \text{ has a solution in } \mathbb{Z}[\omega]$$

is very similar as in the quadratic case (Lemma 1.4) using that  $(\mathbb{Z}[\omega]/\pi\mathbb{Z}[\omega])^*$  is cyclic since it is the multiplicative group of a finite field.

For the other direction, we can use the multiplicativity of the cubic Legendre symbol. In particular, if  $x \in \mathbb{Z}[\omega]$  such that  $x^3 \equiv \alpha \pmod{\pi}$  then

$$\left(\frac{\alpha}{\pi}\right)_3 = \left(\frac{x}{\pi}\right)_3^3 = 1. \quad \square$$

We can also use this to find cubes in  $\mathbb{Z}/p\mathbb{Z}$ .

**Corollary 2.7.** Let  $p \neq 3$  be a prime in  $\mathbb{Z}$ , let  $\alpha \in \mathbb{Z}$  be coprime to  $p$ .

- (i) If  $p \equiv 2 \pmod{3}$  then  $x^3 \equiv \alpha \pmod{p}$  always has a solution in  $\mathbb{Z}$
- (ii) If  $p \equiv 1 \pmod{3}$  then write  $p = \pi\bar{\pi}$  as in Proposition 2.2 and

$$\left(\frac{\alpha}{\pi}\right)_3 = 1 \iff x^3 \equiv \alpha \pmod{p} \text{ has a solution in } \mathbb{Z}.$$

This holds for any prime  $\pi$  satisfying  $p = \pi\bar{\pi}$ .

*Proof.* For the first part notice that  $(\mathbb{Z}/p\mathbb{Z})^*$  is cyclic with  $p - 1$  elements and since  $3 \nmid p - 1$  the map  $a \mapsto a^3$  induces an isomorphism.

For part (ii), let  $\pi$  be a prime in  $\mathbb{Z}[\omega]$  such that  $p = \pi\bar{\pi}$  as in Proposition 2.2, then we get that  $\pi$  is not associate to  $1 - \omega$ . Now by Proposition 2.6

$$\left(\frac{\alpha}{\pi}\right)_3 = 1 \iff x^3 \equiv \alpha \pmod{\pi} \text{ has a solution in } \mathbb{Z}[\omega].$$

We have seen that  $\mathbb{Z}/p\mathbb{Z}$  is naturally isomorphic to  $\mathbb{Z}[\omega]/\pi\mathbb{Z}[\omega]$  (induced by the inclusion  $\mathbb{Z} \hookrightarrow \mathbb{Z}[\omega]$ ) and thus

$$x^3 \equiv \alpha \pmod{p} \text{ has a solution in } \mathbb{Z} \iff x^3 \equiv \alpha \pmod{\pi} \text{ has a solution in } \mathbb{Z}[\omega]. \quad \square$$

To state cubic reciprocity, we need one more definition.

**Definition 2.8.** Let  $\pi$  be a prime in  $\mathbb{Z}[\omega]$ . We call it *primary* if  $\pi \equiv -1 \pmod{3}$ .

**Remark 2.9.** For  $\pi$  be a prime not dividing 3 exactly one of the six associates is primary.

**Example 2.10.** Let  $\pi = 3 + \omega$ , then  $N(\pi) = 7$ , so  $\pi$  is prime. We can find that  $\omega^2\pi$  is the associate prime that is primary, since

$$\omega^2\pi = -3\omega^2 - 1 = 2 - 3(1 - \omega^2) \equiv -1 \pmod{3}.$$

**Theorem 2.11** (Cubic reciprocity, [Cox89], Theorem 4.12). Let  $\pi, \theta$  be primary primes in  $\mathbb{Z}[\omega]$ , coprime to each other. Then

$$\left(\frac{\pi}{\theta}\right)_3 = \left(\frac{\theta}{\pi}\right)_3.$$

**Remark 2.12.** Notice that  $\left(\frac{-1}{\pi}\right)_3 = 1$  for all  $\pi$  prime and not associate to  $1-\omega$ , since  $x^3 \equiv -1 \pmod{\pi}$  always has the obvious solution  $x = -1$ . Furthermore, we also have  $\left(\frac{\theta}{\pi}\right)_3 = \left(\frac{\theta}{-\pi}\right)_3$  by definition of the cubic Legendre symbol. Thus the cubic reciprocity law holds in fact for all  $\pi$ ,  $\theta \equiv \pm 1 \pmod{3}$ .

We will see two proofs of this statement in section 7. To deal with the primes which are not primary, we also have supplementary laws for the cubic Legendre symbol.

**Theorem 2.13** ([Cox89], p.80). Let  $\pi$  be a primary prime in  $\mathbb{Z}[\omega]$  and write  $\pi = -1 + 3m + 3n\omega$ . Then

$$\left(\frac{\omega}{\pi}\right)_3 = \omega^{m+n}, \quad (1)$$

$$\left(\frac{1-\omega}{\pi}\right)_3 = \omega^{2m}. \quad (2)$$

Let us use these theorems to find cubes in residue fields, similar to Section 1.

**Example 2.14.** We want to determine whether 5 is a cube mod 31. First write  $31 = (5 + 6\omega)(\overline{5 + 6\omega})$ . Let  $\pi = 5 + 6\omega$ , then  $\pi$  and 5 are primary and

$$\left(\frac{5}{\pi}\right)_3 = \left(\frac{\pi}{5}\right)_3 = \left(\frac{\omega + 5(1+\omega)}{5}\right)_3 = \left(\frac{\omega}{5}\right)_3 = \omega^2.$$

So 5 is not a cube mod 31 by Corollary 2.7. However,

$$\left(\frac{2}{\pi}\right)_3 = \left(\frac{\pi}{2}\right)_3 = \left(\frac{1 + 2(2 + 3\omega)}{2}\right)_3 = \left(\frac{1}{2}\right)_3 = 1.$$

So 2 is a cube mod 31! In fact,  $4^3 = 64 \equiv 2 \pmod{31}$ .

We can also apply these laws to prove a conjecture of Euler.

**Theorem 2.15.** Let  $p$  be a prime in  $\mathbb{Z}$ . Then

$$4p = x^2 + 243y^2 \text{ has a solution in } \mathbb{Z}^2 \iff p \equiv 1 \pmod{3}, \text{ and } 3 \text{ is a cubic residue mod } p$$

*Proof.* First assume  $p \equiv 1 \pmod{3}$  and 3 is a cubic residue modulo  $p$ . We start by writing  $p = \theta\bar{\theta}$  where we choose  $\theta$  to be primary. Then  $\theta = c + 3d\omega$  with  $c \equiv -1 \pmod{3}$  and

$$4p = 4\theta\bar{\theta} = (2c - 3d)^2 + 27d^2 = a^2 + 27b^2$$

for  $a = 2c - 3d$ ,  $b = d$ , so  $a, b$  have the same parity. The goal is now to show that  $3 \mid b$  if 3 is a cubic residue mod  $p$ .

In order to have easier notation assume from now on that by  $\sqrt{-3}$  we mean  $\sqrt{3}i$ . Notice that  $\sqrt{-3} = 1 + 2\omega$ . Then

$$\pi := \frac{a + 3\sqrt{-3}b}{2} = \frac{a + 3b + 6b\omega}{2} = \frac{a + 3b}{2} + 3b\omega \in \mathbb{Z}[\pi]$$

since  $a + 3b$ ,  $6b$  are divisible by 2. We claim  $\pi$  is a primary prime.

We have

$$N(\pi) = \pi\bar{\pi} = \frac{a + 3\sqrt{-3}b}{2} \cdot \frac{a - 3\sqrt{-3}b}{2} = \frac{a^2 + 27b^2}{4} = p,$$

thus  $\pi$  is prime in  $\mathbb{Z}[\omega]$ . Moreover, since  $\theta$  was primary, so  $\frac{a+3b}{2} = c \equiv -1 \pmod{3}$  and  $\pi$  is again primary.

Now we need to use the supplementary laws. Notice that from (1) and (2), if we write  $\pi = -1 + 3m + 3b\omega$  ( $\pi$  is primary), we have using that  $3 = -\omega^2(1 - \omega)^2$

$$\left(\frac{3}{\pi}\right)_3 = \left(\frac{-1}{\pi}\right)_3 \cdot \left(\frac{\omega}{\pi}\right)_3^2 \cdot \left(\frac{1 - \omega}{\pi}\right)_3^2 = \omega^{2m+2b} \cdot \omega^{4m} = \omega^{2b}.$$

We used that  $\left(\frac{-1}{\pi}\right)_3 = 1$  by Remark 2.12.

Using now that 3 is a cubic residue mod  $p$ , by Corollary 2.7

$$\left(\frac{3}{\pi}\right)_3 = \omega^{2b} = 1.$$

But this implies  $3 \mid b$  and

$$4p = a^2 + 27b^2 = x^2 + 243y^2$$

for  $x = a$ ,  $y = \frac{b}{3}$ .

To show the other direction, let  $x, y \in \mathbb{Z}$  such that

$$4p = x^2 + 243y^2.$$

Reducing modulo 3 we see that  $p \equiv 1 \pmod{3}$ , since the only squares modulo 3 are 0 and 1. Clearly since  $p$  is prime, it cannot be 0 modulo 3, unless  $p = 3$  but then we can easily see that the equation has no integer solution. Similarly to above, now set

$$\pi := \frac{x + 9\sqrt{-3}y}{2} = \frac{x + 9y}{2} + 9y\omega,$$

then  $N(\pi) = \pi\bar{\pi} = p$  so  $\pi$  is a prime in  $\mathbb{Z}[\omega]$ . Also notice that either  $\pi$  or  $-\pi$  is primary. We now deduce

$$\begin{cases} \left(\frac{3}{\pi}\right)_3 = \omega^{6y} = 1 & \text{if } \pi \text{ is primary, and} \\ \left(\frac{3}{-\pi}\right)_3 = \omega^{-6y} = 1 & \text{if } -\pi \text{ is primary.} \end{cases}$$

Thus 3 is a cubic residue mod  $p$  by Corollary 2.7. □

**Remark 2.16.** A much more well-known conjecture of Euler is

$$p = x^2 + 27y^2 \iff \begin{cases} p \equiv 1 \pmod{3}, \text{ and} \\ 2 \text{ is a cubic residue mod } p \end{cases}$$

We can prove it in a very similar way using cubic reciprocity. For details, we refer to ([Cox89], Theorem 4.15).

### 3 Ramification in Galois Field Extensions

In this section we want to develop some tools to help us understand the theorems of class field theory. It is mainly based on Marcus' book 'Number Fields' [Mar18]. We begin by repeating a couple of results from algebraic number theory.

Let  $K \subset L$  be number fields. We denote by  $O_K$  the ring of algebraic integers in  $K$ . We have the norm given by  $N_{L/K}(\alpha) = \prod_{\sigma \in \text{Gal}(L/K)} \sigma(\alpha)$  for  $\alpha \in L$ . For a prime ideal  $\mathfrak{p}$  in  $O_K$ , the norm  $N(\mathfrak{p})$  is given by  $N(\mathfrak{p}) = |O_K/\mathfrak{p}|$ . From now on, we will write prime instead of primes ideals, unless there might be some confusion.

We know that for every prime  $\mathfrak{p}$  in  $O_K$ , there exist prime ideals  $\mathfrak{P}_1, \dots, \mathfrak{P}_n$  in  $O_L$  such that  $\mathfrak{p}O_K = \mathfrak{P}_1^{e_1} \cdots \mathfrak{P}_n^{e_n}$ . These are exactly the prime ideals in  $O_L$  containing  $\mathfrak{p}$ , and we have  $\mathfrak{P}_i \cap O_K = \mathfrak{p}$ . We say the  $\mathfrak{P}_i$ 's lie over  $\mathfrak{p}$ .

For  $\mathfrak{P}$  lying over  $\mathfrak{p}$ , we call the corresponding exponent in the prime decomposition the *ramification index*, denoted by  $e(\mathfrak{P}|\mathfrak{p})$ . Furthermore, the inclusion  $K \hookrightarrow L$  induces an embedding  $O_K/\mathfrak{p} \hookrightarrow O_L/\mathfrak{P}$ . Since both residue fields are finite we have an extension of finite fields. We call the degree of that field extension the *inertial degree*, denoted by  $f(\mathfrak{P}|\mathfrak{p})$ .

Let  $\mathfrak{p}O_K = \mathfrak{P}_1^{e_1} \cdots \mathfrak{P}_r^{e_r}$ . If  $f_i$  are the corresponding inertial degrees, then  $\sum_{i=1}^r e_i f_i = [L : K]$ .

- (i) We say  $\mathfrak{p}$  is *ramified* if  $e_i \geq 2$  for some  $i$ .
- (ii) We say  $\mathfrak{p}$  is *totally ramified* if  $\mathfrak{p}O_K = \mathfrak{P}^d$  for  $d = [L : K]$ .
- (iii) We say  $\mathfrak{p}$  *splits completely* if  $\mathfrak{p}O_K$  splits into  $d$  distinct prime ideals.

**Remark 3.1.** If  $K \subset M \subset L$  is a tower of fields,  $\mathfrak{p}$  a prime in  $K$ ,  $\mathfrak{P}$  a prime in  $M$  lying over  $\mathfrak{p}$  and  $\mathfrak{P}'$  a prime in  $L$  lying over  $\mathfrak{P}$ . Then  $e(\mathfrak{P}'|\mathfrak{p}) = e(\mathfrak{P}'|\mathfrak{P})e(\mathfrak{P}|\mathfrak{p})$  and  $f(\mathfrak{P}'|\mathfrak{p}) = f(\mathfrak{P}'|\mathfrak{P})f(\mathfrak{P}|\mathfrak{p})$ .

Now we look at how for  $\mathfrak{p}$  a prime in  $K$  the Galois group of a normal extension acts on the prime ideals lying over  $\mathfrak{p}$ .

**Theorem 3.2** ([Mar18], Theorem 23). Let  $K \subset L$  be a normal extension,  $\mathfrak{P}, \mathfrak{P}'$  two primes of  $O_L$  lying over the same prime  $\mathfrak{p}$  of  $O_K$ . Then  $\sigma(\mathfrak{P}) = \mathfrak{P}'$  for some  $\sigma \in \text{Gal}(L/K)$ .

*Proof.* Let  $\mathfrak{P}, \mathfrak{P}'$  be two primes of  $O_L$  lying over the same prime  $\mathfrak{p}$  of  $O_K$ . First, we note that if  $\sigma \in \text{Gal}(L/K)$  then  $\sigma(\mathfrak{P})$  is again a prime of  $O_L$  lying over  $\mathfrak{p}$  since  $\mathfrak{p}$  must be fixed by the Galois group. Suppose  $\sigma(\mathfrak{P}) \neq \mathfrak{P}'$  for all  $\sigma \in \text{Gal}(L/K)$ . By the Chinese remainder theorem, we can choose  $\alpha \in O_L$  such that

$$\begin{aligned} \alpha &\equiv 0 \pmod{\mathfrak{P}'}, \\ \alpha &\equiv 1 \pmod{\sigma(\mathfrak{P})} \quad \forall \sigma \in \text{Gal}(L/K). \end{aligned}$$

But now notice that the norm  $N_{L/K}(\alpha) = \prod_{\sigma \in \text{Gal}(L/K)} \sigma(\alpha)$  lies in  $O_L \cap K = O_K$  (it lies in  $K$  since it is the constant coefficient of  $\alpha$ 's minimal polynomial) and in  $\mathfrak{P}'$  since  $\alpha \in \mathfrak{P}'$ . Thus  $N_{L/K}(\alpha)$  lies in  $O_K \cap \mathfrak{P}' = \mathfrak{p}$ .

But since  $\sigma^{-1}(\alpha) \notin \mathfrak{P}$  for all  $\sigma \in \text{Gal}(L/K)$  and

$$N_{L/K}(\alpha) = \prod_{\sigma \in \text{Gal}(L/K)} \sigma(\alpha) = \prod_{\sigma \in \text{Gal}(L/K)} \sigma^{-1}(\alpha),$$

we can deduce  $N_{L/K}(\alpha) \notin \mathfrak{P}$ . But this is clearly a contradiction to  $N_{L/K}(\alpha) \in \mathfrak{p}$ .  $\square$

**Corollary 3.3.** Let  $\mathfrak{P}, \mathfrak{P}'$  be two primes of  $O_L$  lying over the same prime  $\mathfrak{p}$  of  $O_K$ . Then  $e(\mathfrak{P}|\mathfrak{p}) = e(\mathfrak{P}'|\mathfrak{p})$  and  $f(\mathfrak{P}|\mathfrak{p}) = f(\mathfrak{P}'|\mathfrak{p})$ .

*Proof.*  $e(\mathfrak{P}|\mathfrak{p}) = e(\mathfrak{P}'|\mathfrak{p})$  follows from the uniqueness of the prime decomposition of  $\mathfrak{p}$  and the fact that the Galois group permutes the primes over  $\mathfrak{p}$ . The second part follows since  $\sigma \in \text{Gal}(L/K)$  induces an isomorphism  $O_L/\mathfrak{P} \rightarrow O_L/\mathfrak{P}'$  if  $\sigma(\mathfrak{P}) = \mathfrak{P}'$ .  $\square$

From now on let  $K \subset L$  be Galois,  $\mathfrak{p}$  a prime in  $O_K$  and  $\mathfrak{P}$  a prime in  $O_L$  lying over  $\mathfrak{p}$ . Again, if  $\mathfrak{p}$  splits into  $r$  primes with ramification indices  $e$  and inertial degrees  $f$  then  $ref = [L : K]$ . Notice here that we used that all primes over  $\mathfrak{p}$  have the same ramification indices and inertial degrees.

**Definition 3.4.** We define the *decomposition group* to be

$$D(\mathfrak{P}|\mathfrak{p}) = \{\sigma \in \text{Gal}(L/K) : \sigma(\mathfrak{P}) = \mathfrak{P}\}.$$

Next, we define the *inertia group* to be

$$I(\mathfrak{P}|\mathfrak{p}) = \{\sigma \in \text{Gal}(L/K) : \sigma(\alpha) \equiv \alpha \pmod{\mathfrak{P}} \forall \alpha \in O_L\}.$$

It is trivial to see that  $I(\mathfrak{P}|\mathfrak{p})$  is a normal subgroup of  $D(\mathfrak{P}|\mathfrak{p})$ .

**Remark 3.5.** The decomposition group is isomorphic to the Galois group of the field extension of local fields  $K_{\mathfrak{p}} \subset L_{\mathfrak{P}}$ , where  $K_{\mathfrak{p}}, L_{\mathfrak{P}}$  are the completions of  $K, L$  at  $\mathfrak{p}, \mathfrak{P}$ .

**Theorem 3.6.** Let  $\mathfrak{P}$  be one of the  $r$  primes lying over  $\mathfrak{p}$  with ramification index  $e$  and inertial degree  $f$ . Moreover, let  $D = D(\mathfrak{P}|\mathfrak{p})$ ,  $I = I(\mathfrak{P}|\mathfrak{p})$ , let  $L^D, L^I$  be the fixed fields of  $D, I$  and finally let  $\mathfrak{P}_D = L^D \cap \mathfrak{P}$  and  $\mathfrak{P}_I = L^I \cap \mathfrak{P}$ . Then we have the following diagram

| degrees | $L$   | $\mathfrak{P}$   | ramification indices | inertial degrees |
|---------|-------|------------------|----------------------|------------------|
| $e$     | $L^I$ | $\mathfrak{P}_I$ | $e$                  | 1                |
| $f$     | $L^D$ | $\mathfrak{P}_D$ | 1                    | $f$              |
| $r$     | $K$   | $\mathfrak{p}$   | 1                    | 1                |

*Proof.* (i) We start by showing  $[L^D : K] = r$ . Note that  $\text{Gal}(L/K)$  acts transitively on the set of all primes lying over  $\mathfrak{p}$  by Theorem 1.1. It is easy to see that the stabilizer of  $\mathfrak{P}$  is  $D$  and

thus by orbit-stabilizer theorem we have  $[\text{Gal}(L/K) : D] = r$ . By Galois correspondence we deduce that  $[L^D : K] = r$ .

Since  $\text{Gal}(L/L^D) = D$  stabilizes  $\mathfrak{P}$ , by Proposition 3.2  $\mathfrak{P}$  is the only prime lying over  $\mathfrak{P}_D$  in  $L$ . We therefore have

$$e(\mathfrak{P}|\mathfrak{P}_D)f(\mathfrak{P}|\mathfrak{P}_D) = [L : L^D] = \frac{[L : K]}{[L^D : K]} = ef$$

but since  $e(\mathfrak{P}|\mathfrak{P}_D) \leq e$ ,  $f(\mathfrak{P}|\mathfrak{P}_D) \leq f$  we deduce  $e(\mathfrak{P}|\mathfrak{P}_D) = e$ ,  $f(\mathfrak{P}|\mathfrak{P}_D) = f$ . Thus by multiplicativity of ramification indices and inertial degrees,  $e(\mathfrak{P}_D|\mathfrak{p}) = f(\mathfrak{P}_D|\mathfrak{p}) = 1$ .

- (ii) Next we show  $f(\mathfrak{P}|\mathfrak{P}_I) = 1$  or equivalently that the field extension of residue fields is trivial. For an element  $\theta \in O_L/\mathfrak{P}$  choose a lift  $\alpha \in O_L$ . The polynomial

$$\prod_{\sigma \in I} (X - \sigma(\alpha))$$

has coefficients in  $O_{L^I}$  since it is the minimal polynomial of  $\alpha$ . By definition of the inertia group  $\sigma(\alpha) \equiv \alpha \pmod{\mathfrak{P}}$  so in  $O_L/\mathfrak{P}$  the polynomial reduces to

$$\prod_{\sigma \in I} (X - \theta)$$

with coefficients in  $O_{L^I}/\mathfrak{P}_I$ . But  $\text{Gal}((O_L/\mathfrak{P})/(O_{L^I}/\mathfrak{P}_I))$  can only permute zeros of any polynomial with coefficients in  $O_{L^I}/\mathfrak{P}_I$ . In particular,  $\text{Gal}((O_L/\mathfrak{P})/(O_{L^I}/\mathfrak{P}_I))$  fixes  $\theta$ . Since  $\theta$  was arbitrary we see that the Galois group is trivial, showing  $f(\mathfrak{P}|\mathfrak{P}_I) = 1$ .

- (iii) We can directly deduce that  $f(\mathfrak{P}_I|\mathfrak{P}_D) = f$  by multiplicativity of inertial degrees. Furthermore, notice that  $[D : I] \leq f$  since we can map  $D$  into the Galois group of  $O_K/\mathfrak{p} \subset O_L/\mathfrak{P}$  via reduction modulo  $\mathfrak{P}$  and it is easy to see then the inertia group is exactly the kernel. Since  $[L^I : L^D] \geq f(\mathfrak{P}_I|\mathfrak{P}_D) = f$  we get  $[D : I] = [L^I : L^D] = f$  and the rest follows from the multiplicativity of the degrees.  $\square$

**Proposition 3.7** ([Mar18], Theorem 29). With all notation as in Theorem 3.6 we have that

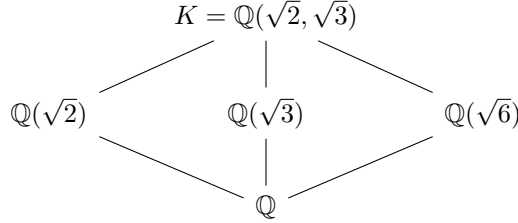
- (i)  $L_D$  is the largest intermediate field  $K'$  such that for  $\mathfrak{P}' = K' \cap \mathfrak{P}$ ,  $e(\mathfrak{P}'|\mathfrak{p}) = f(\mathfrak{P}'|\mathfrak{p}) = 1$ .
- (ii)  $L_I$  is the largest intermediate field  $K'$  such that  $e(\mathfrak{P}'|\mathfrak{p}) = 1$ .

In particular the inertia group is trivial if  $\mathfrak{p}$  is unramified in  $L$ .

*Proof.* See Marcus ([Mar18], p.73).  $\square$

**Remark 3.8.** If  $D$  is a normal subgroup then  $L_D$  is the largest intermediate field such that  $\mathfrak{p}$  splits completely in it. Similarly, if  $I$  is a normal subgroup then  $L_I$  is the largest intermediate field such that  $\mathfrak{p}$  is unramified in it. In particular this implies that in abelian extensions there is a unique largest field in which a given prime splits or is unramified. This can help simplify some computations in the future.

**Example 3.9.** Regard the quartic field extension  $K = \mathbb{Q}(\sqrt{2}, \sqrt{3})$ . The extension  $\mathbb{Q} \subset K$  is clearly Galois with  $\text{Gal}(K/\mathbb{Q}) \cong C_2 \times C_2$ . We can write  $K = \mathbb{Q}(\sqrt{2} + \sqrt{3})$ , with the minimal polynomial of  $\sqrt{2} + \sqrt{3}$  being  $X^4 - 10X^2 + 1$ .



- (i)  $\langle 5 \rangle$  is unramified in  $K$ . It is inert in  $\mathbb{Q} \subset \mathbb{Q}(\sqrt{2})$  and splits in  $\mathbb{Q} \subset \mathbb{Q}(\sqrt{6})$ . We can deduce that the fixed field of the decomposition group for every prime over 5 is  $\mathbb{Q}(\sqrt{6})$ , thus the decomposition group for every prime over 5 is  $\text{Gal}(K/\mathbb{Q}(\sqrt{6}))$ . The fixed field of the inertia group is  $K$ , thus the inertia group is trivial.
- (ii)  $\langle 2 \rangle$  ramifies in  $\mathbb{Q} \subset \mathbb{Q}(\sqrt{2})$ ,  $\mathbb{Q} \subset \mathbb{Q}(\sqrt{3})$ , and also in  $\mathbb{Q} \subset \mathbb{Q}(\sqrt{6})$ . We can directly deduce that the fixed fields of the decomposition group and inertia group are both  $\mathbb{Q}$ , so the decomposition group and the inertia group of every prime over 2 are both the Galois group  $\text{Gal}(K/\mathbb{Q})$ .
- (iii)  $\langle 3 \rangle$  is inert in  $\mathbb{Q} \subset \mathbb{Q}(\sqrt{2})$  and ramifies in  $\mathbb{Q} \subset \mathbb{Q}(\sqrt{3})$ . We can deduce that in  $K$ ,  $\langle 3 \rangle = \mathfrak{p}^2$  where  $\mathfrak{p}$  is a prime in  $K$  and has inertial degree 2. Furthermore, the fixed field of the decomposition group is  $\mathbb{Q}$ , thus  $D(\mathfrak{p}|\langle 3 \rangle) = \text{Gal}(K/\mathbb{Q})$ , and the fixed field of the inertia group is  $\mathbb{Q}(\sqrt{2})$ , thus  $I(\mathfrak{p}|\langle 3 \rangle) = \text{Gal}(K/\mathbb{Q}(\sqrt{2}))$ .

Notice that we know that  $\langle 3 \rangle$  ramifies in  $\mathbb{Q} \subset \mathbb{Q}(\sqrt{6})$  without checking because Remark 3.8 tells us that there is a unique largest field such that 3 is unramified in it.

Let  $K \subset L$ , let  $\mathfrak{p}$  be a prime in  $K$  and let  $\mathfrak{P}$  be a prime over  $\mathfrak{p}$  in  $L$ . Let  $\overline{G}$  be the Galois group of the field extension  $O_K/\mathfrak{p} \subset O_L/\mathfrak{P}$ . We have a group homomorphism  $D(\mathfrak{P}|\mathfrak{p}) \rightarrow \overline{G}$  with kernel  $I(\mathfrak{P}|\mathfrak{p})$ . Using that  $[D : I] = f = |(O_L/\mathfrak{P})/(O_K/\mathfrak{p})|$  from Theorem 3.6, we can deduce that  $D(\mathfrak{P}|\mathfrak{p}) \rightarrow \overline{G}$  is surjective. Thus we have

$$D(\mathfrak{P}|\mathfrak{p})/I(\mathfrak{P}|\mathfrak{p}) \cong \overline{G}.$$

Since the residue fields  $O_K/\mathfrak{p}$  and  $O_L/\mathfrak{P}$  are finite, the Galois group of the field extension is cyclic. Furthermore, it is generated by the Frobenius homomorphism  $\phi \in \overline{G}$  with

$$\phi(\alpha) = \alpha^{N(\mathfrak{p})} \text{ for every } \alpha \in O_L/\mathfrak{P}$$

where  $N(\mathfrak{p}) = |O_K/\mathfrak{p}|$ .

**Proposition 3.10** ([Mar18], Theorem 32). If  $\mathfrak{p}$  is unramified then there is a unique element  $\sigma \in \text{Gal}(L/K)$  that satisfies

$$\sigma(\alpha) \equiv \alpha^{N(\mathfrak{p})} \pmod{\mathfrak{P}} \text{ for every } \alpha \in O_L.$$

*Proof.* We have  $D(\mathfrak{P}|\mathfrak{p})/I(\mathfrak{P}|\mathfrak{p}) \cong \overline{G}$  but since  $\mathfrak{p}$  is unramified,  $I(\mathfrak{P}|\mathfrak{p})$  is trivial. Thus there is a unique element in  $D(\mathfrak{P}|\mathfrak{p})$  that maps to the Frobenius homomorphism in  $\overline{G}$ .  $\square$

## 4 Theorems of Class Field Theory and the Artin Map

About a hundred years after Euler found his quadratic reciprocity law, Emil Artin discovered the up until then most general reciprocity law. In this section we want to state and understand the main theorems in the ideal-theoretic formulation of global class field theory. This part mostly follows Chapter 2 of Cox's book 'Primes of the form  $x^2 + ny^2$ ' ([Cox89]).

In the classical version, class field theory deals with the description of all abelian field extensions of a given ground field. From now on, we will mostly work with abelian extensions. Recall that we call a Galois field extension  $K \subset L$  abelian if the Galois group  $\text{Gal}(L/K)$  is abelian.

Throughout this section let  $K \subset L$  be number fields, let  $\mathfrak{p}$  be a prime in  $K$ , and let  $\mathfrak{P}$  be a prime in  $L$  lying over  $\mathfrak{p}$ . Furthermore assume  $\mathfrak{p}$  is unramified in  $L$ .

**Definition 4.1.** We define the Artin symbol  $\left(\frac{L/K}{\mathfrak{P}}\right)$  to be the unique element in  $\text{Gal}(L/K)$  from Proposition 3.10, i.e.,

$$\left(\frac{L/K}{\mathfrak{P}}\right)(\alpha) \equiv \alpha^{N(\mathfrak{P})} \pmod{\mathfrak{P}} \quad \text{for every } \alpha \in O_L.$$

**Proposition 4.2** ([Cox89], Cor.5.21). If  $\sigma \in \text{Gal}(L/K)$ , then

$$\left(\frac{L/K}{\sigma(\mathfrak{P})}\right) = \sigma \left(\frac{L/K}{\mathfrak{P}}\right) \sigma^{-1}$$

*Proof.* We have that for all  $\alpha \in O_L$ ,  $\left(\frac{L/K}{\mathfrak{P}}\right)\sigma^{-1}(\alpha) \equiv \sigma^{-1}(\alpha) \pmod{\mathfrak{P}}$ . Thus  $\sigma\left(\frac{L/K}{\mathfrak{P}}\right)\sigma^{-1}(\alpha) \equiv \alpha \pmod{\sigma(\mathfrak{P})}$ . By uniqueness of the Artin map the equality follows.  $\square$

In abelian extensions the Artin map will thus only depend on  $\mathfrak{p}$ . We will thus proceed by just writing  $\left(\frac{L/K}{\mathfrak{p}}\right)$  for the Artin map.

Remember that we have used the word prime for prime ideals, which we will also call *finite* primes. We now introduce *infinite primes*. A *real* infinite prime will be given by an embedding  $K \hookrightarrow \mathbb{R}$  and a *complex* infinite prime will be given by a pair of embeddings  $\sigma, \bar{\sigma} : K \hookrightarrow \mathbb{C}$ . We also have a notion of ramification for infinite primes. A real infinite prime of  $K$ , i.e., a real embedding, ramifies in  $L$  if it can be extended to a complex embedding  $L \hookrightarrow \mathbb{C}$ .

**Remark 4.3.** The concept of infinite primes might seem a bit odd at first. But if we look at *places* of a number field  $K$ , i.e. equivalence classes of absolute values on  $K$ , the notion of finite and infinite primes seems more natural. In fact, the nonarchimedean absolute values correspond exactly to the finite primes of  $K$  and the other absolute values correspond to the infinite primes of  $K$ .

As an easy example for  $K = \mathbb{Q}$  we have the following places:

- (i) the  $p$ -adic absolute values  $|\cdot|_p$  corresponding to the finite primes  $p \in \mathbb{Z}$ ,
- (ii) the euclidean absolute value  $|\cdot|_\infty$  corresponding to the embedding  $\mathbb{Q} \hookrightarrow \mathbb{R}$ .

Since not all abelian extensions of number fields are unramified, we introduce the concept of a modulus. One can think of a modulus as a product of (ramified) primes that we want to avoid.

**Definition 4.4.** A modulus is a formal product

$$\mathfrak{m} = \prod_{\mathfrak{p}} \mathfrak{p}^{n_{\mathfrak{p}}}$$

over all primes (finite or infinite) where only finitely many exponents are nonzero and for infinite primes

$$n_{\mathfrak{p}} = \begin{cases} 0 & \text{if } \mathfrak{p} \text{ is a complex embedding} \\ 0 \text{ or } 1 & \text{if } \mathfrak{p} \text{ is a real embedding} \end{cases}$$

We will mostly consider a modulus  $\mathfrak{m}$  as a product of an ideal  $m_0$  and a product of real infinite primes  $m_\infty$ . If all exponents are zero we set  $\mathfrak{m} = 1$ .

For a given modulus  $\mathfrak{m} = m_0 m_\infty$  let  $I_K(\mathfrak{m})$  be the group of all fractional ideals that are relatively prime to  $m_0$ . Furthermore, let  $P_{K,1}(\mathfrak{m})$  be the subgroup of  $I_K(\mathfrak{m})$  generated by all principal ideals  $\alpha O_K$  such that  $\alpha \equiv 1 \pmod{m_0}$  and  $\sigma(\alpha) > 0$  for all  $\sigma$  dividing  $m_\infty$ .

Now let  $K \subset L$  be an abelian field extension and  $\mathfrak{m}$  a modulus in  $K$  divisible by all primes which ramify. Note that only finitely many primes ramify in  $K \subset L$ . We can extend our Artin symbol that currently is defined for unramified prime ideals by multiplicativity to a map

$$\Phi_{\mathfrak{m}} : I_K(\mathfrak{m}) \longrightarrow \text{Gal}(L/K).$$

We will call this the Artin map of  $K \subset L$  and write  $\Phi_{L/K, \mathfrak{m}}$  if it is unclear what field extension we are working with.

**Definition 4.5.** For a modulus  $\mathfrak{m}$  in  $K$ , we call a subgroup  $H \subset I_K(\mathfrak{m})$  a *congruence subgroup* for  $\mathfrak{m}$  if  $P_{K,1}(\mathfrak{m}) \subset H$ . If  $H$  is a congruence subgroup then we call  $I_K(\mathfrak{m})/H$  a *generalized ideal class group*.

**Remark 4.6.** Note that for  $\mathfrak{m} = 1$ ,  $I_K(\mathfrak{m})/P_{K,1}(\mathfrak{m}) = I_K/P_K$  is the ideal class group.

In the following we will state the main theorems of class field theory - without proof. For readers interested in the proofs we refer to Milne's notes on class field theory [Mil20].

**Theorem 4.7** (Artin reciprocity, [Cox89], Theorem 8.2). Let  $K \subset L$  be an abelian extension and  $\mathfrak{m}$  a modulus divisible by all primes of  $K$  that ramify. Then

- (i) The Artin map  $\Phi_{\mathfrak{m}}$  is surjective and thus

$$I_K(\mathfrak{m})/\ker(\Phi_{\mathfrak{m}}) \xrightarrow{\sim} \text{Gal}(L/K).$$

- (ii) If the exponents of the modulus are sufficiently large then  $\ker(\Phi_{\mathfrak{m}})$  is a congruence subgroup for  $\mathfrak{m}$ . Consequently,  $\text{Gal}(L/K)$  is a generalized ideal class group.

The next theorem asserts that there is always a best possible modulus.

**Theorem 4.8** (Conductor Theorem, [Cox89], Theorem 8.5). Let  $K \subset L$  be an abelian extension. Then there is a modulus  $\mathfrak{f} = \mathfrak{f}(L/K)$  such that

- (i) A prime of  $K$  ramifies if and only if it divides  $\mathfrak{f}$ .  
 (ii) If  $\mathfrak{m}$  is any modulus divisible by all primes of  $K$  that ramify then  $\Phi_{\mathfrak{m}}$  is a congruence subgroup for  $\mathfrak{m}$  if and only if  $\mathfrak{f}$  divides  $\mathfrak{m}$ .

**Definition 4.9.** We will call  $\mathfrak{f}$  as given in the Theorem the *conductor* of  $L/K$ .

**Theorem 4.10** (Existence Theorem, [Cox89], Theorem 8.6). Let  $K$  be a number field,  $\mathfrak{m}$  a modulus in  $K$  and  $H$  a congruence subgroup for  $\mathfrak{m}$ . Then there exists a unique abelian field extension  $L$  of  $K$  such that  $\mathfrak{m}$  is divisible by all primes of  $K$  that ramify in  $L$  and the Artin map  $\Phi_{\mathfrak{m}}$  satisfies

$$H = \ker(\Phi_{\mathfrak{m}}).$$

**Definition 4.11.** For a given modulus  $\mathfrak{m}$ , we will call the field extension  $K_{\mathfrak{m}}$  given by Theorem 4.10 with  $H = P_{K,1}(\mathfrak{m})$  the *ray class field* of  $\mathfrak{m}$ .

**Corollary 4.12** ([Cox89], Theorem 8.7). Let  $L, M$  be abelian extensions of  $K$ . Then  $L \subset M$  if and only if there is a modulus  $\mathfrak{m}$ , divisible by all primes that ramify in either  $L$  or  $M$ , such that

$$P_{K,1}(\mathfrak{m}) \subset \ker(\Phi_{M/K,\mathfrak{m}}) \subset \ker(\Phi_{L/K,\mathfrak{m}}).$$

*Proof.* First, we assume that  $L \subset M$ . By Artin reciprocity we can find a modulus  $\mathfrak{m}$  such that  $\ker(\Phi_{M/K,\mathfrak{m}})$  and  $\ker(\Phi_{L/K,\mathfrak{m}})$  are congruence subgroups (to be precise, we can find separate moduli but we can take their product). Let  $r : \text{Gal}(M/K) \rightarrow \text{Gal}(L/K)$  be the restriction map, which is well-defined since  $M, L$  are both Galois extensions. We show that  $\Phi_{L/K,\mathfrak{m}} = r \circ \Phi_{M/K,\mathfrak{m}}$ . It suffices to check the equality on primes. Let  $\mathfrak{p}$  be a prime in  $O_K$ ,  $\mathfrak{P}$  lying over  $\mathfrak{p}$  in  $O_L$  and  $\mathfrak{P}'$  lying over  $\mathfrak{p}$  in  $O_M$  and then for every  $\alpha \in O_L$

$$\left(\frac{M/K}{\mathfrak{p}}\right)(\alpha) \equiv \alpha^{N(\mathfrak{p})} \pmod{\mathfrak{P}'}.$$

Now since  $\mathfrak{P}' \cap O_L = \mathfrak{P}$  and  $\left(\frac{M/K}{\mathfrak{p}}\right)(\alpha), \alpha^{N(\mathfrak{p})} \in O_L$  thus

$$\left(\frac{M/K}{\mathfrak{p}}\right)(\alpha) \equiv \alpha^{N(\mathfrak{p})} \pmod{\mathfrak{P}}$$

in  $O_L$ . By uniqueness of the Artin symbol we deduce  $\Phi_{L/K,\mathfrak{m}} = r \circ \Phi_{M/K,\mathfrak{m}}$  and thus  $\ker(\Phi_{M/K,\mathfrak{m}}) \subset \ker(\Phi_{L/K,\mathfrak{m}})$ .

Now assume  $\mathfrak{m}$  is a modulus, divisible by all primes that ramify in either  $L$  or  $M$ , such that

$$P_{K,1}(\mathfrak{m}) \subset \ker(\Phi_{M/K,\mathfrak{m}}) \subset \ker(\Phi_{L/K,\mathfrak{m}}).$$

The image of  $\ker(\Phi_{L/K, \mathfrak{m}})$  under  $\Phi_{M/K, \mathfrak{m}}$  is a subgroup  $H \subset \text{Gal}(M/K)$ . Denote its fixed field by  $L^H$ , then  $K \subset L^H \subset M$ . By the first part, the Artin map  $\Phi_{L^H/K, \mathfrak{m}}$  is just  $r \circ \Phi_{M/K, \mathfrak{m}}$ , for  $r : \text{Gal}(M/K) \rightarrow \text{Gal}(L^H/K)$  the restriction map. Thus

$$\ker(\Phi_{L^H/K, \mathfrak{m}}) = \ker(r \circ \Phi_{M/K, \mathfrak{m}}) = \Phi_{M/K, \mathfrak{m}}^{-1}(\ker(r)) = \Phi_{M/K, \mathfrak{m}}^{-1}(H) = \ker(\Phi_{L/K, \mathfrak{m}})$$

where we used that  $\ker(r) = \text{Gal}(M/L^H) = H$ . By the uniqueness part of the Existence Theorem 4.10 we get  $L^H = L$  and thus  $L \subset M$ .  $\square$

In this proof we have also shown the following fact.

**Proposition 4.13.** Let  $K \subset M$  be an abelian field extension and let  $K \subset L \subset M$  be an intermediate field. For a modulus  $\mathfrak{m}$ , divisible by all primes that ramify in  $M$  (and thus all the primes that ramify in  $L$ ), and the restriction map

$$\begin{aligned} r : \text{Gal}(M/K) &\longrightarrow \text{Gal}(L/K) \\ \sigma &\mapsto \sigma|_L \end{aligned}$$

we have

$$\Phi_{L/K, \mathfrak{m}} = r \circ \Phi_{M/K, \mathfrak{m}}.$$

To illustrate the statements of these theorems we will take a closer look at the example of cyclotomic field extensions of  $\mathbb{Q}$ .

**Example 4.14.** Let  $\zeta_m = e^{\frac{2\pi i}{m}}$  be a primitive  $m$ -th root of unity. We want to study the Artin map of the abelian field extension  $\mathbb{Q} \subset \mathbb{Q}(\zeta_m)$ .

- (i) We start by finding a suitable modulus for the extension. If  $\infty$  is the real embedding of  $\mathbb{Q}$  into  $\mathbb{C}$  then  $\mathfrak{m} = m\infty$  is divisible by all primes of  $\mathbb{Q}$  that ramify. This follows from Dedekind's criterion and the fact that the minimal polynomial of  $\zeta_m$  divides  $X^m - 1$  which is separable modulo all primes of  $\mathbb{Q}$  except possibly the ones that divide  $m$ . Thus the only primes that possibly ramify are the ones dividing  $m$ .

We can now regard the Artin map

$$\Phi_{\mathfrak{m}} : I_{\mathbb{Q}}(\mathfrak{m}) \longrightarrow \text{Gal}(L/\mathbb{Q}) \cong (\mathbb{Z}/m\mathbb{Z})^*$$

where we identify  $\text{Gal}(L/\mathbb{Q}) \cong (\mathbb{Z}/m\mathbb{Z})^*$  via  $\sigma \mapsto k$ , such that  $\sigma(\zeta_m) = \zeta_m^k$ .

First let  $p$  be a prime of  $\mathbb{Z}$  not dividing  $m$ . Then  $\Phi_{\mathfrak{m}}(p\mathbb{Z})(\zeta_m) \equiv \zeta_m^p \pmod{\mathfrak{P}}$  if  $\mathfrak{P}$  lies over  $p$ . Now since  $p$  is prime to  $m$ , the map  $\zeta_m \mapsto \zeta_m^p$  is an element of the Galois group  $\text{Gal}(L/\mathbb{Q})$  and by uniqueness this is exactly  $\Phi_{\mathfrak{m}}(p\mathbb{Z})$ , i.e.  $\Phi_{\mathfrak{m}}(p\mathbb{Z}) = [p] \in (\mathbb{Z}/m\mathbb{Z})^*$ .

Since  $\mathbb{Z}$  is a principal ideal domain, all elements of  $I_{\mathbb{Q}}(\mathfrak{m})$  take the form  $\frac{a}{b}\mathbb{Z}$ , where  $a, b > 0$  are coprime to  $m$ . So if we extend the Artin Map by multiplicativity we get a description

$$\Phi_{\mathfrak{m}}\left(\frac{a}{b}\mathbb{Z}\right) = [a][b]^{-1} \in (\mathbb{Z}/m\mathbb{Z})^*$$

for  $a, b > 0$  coprime to  $m$ .

Next, let us determine the kernel of  $\Phi_{\mathfrak{m}}$ . If  $a, b > 0$ , coprime to  $m$  and  $\Phi_{\mathfrak{m}}(\frac{a}{b}\mathbb{Z}) = 1 \in (\mathbb{Z}/m\mathbb{Z})^*$  then  $a \equiv b \pmod{m}$ . We can choose  $u \in \mathbb{Z}$ ,  $u > 0$  such that  $au, bu \equiv 1 \pmod{m}$ . Then  $\frac{a}{b}\mathbb{Z} = au\mathbb{Z} \cdot (bu\mathbb{Z})^{-1} \in P_{\mathbb{Q},1}(\mathfrak{m})$ . So we get  $\ker(\Phi_{\mathfrak{m}}) \subset P_{\mathbb{Q},1}(\mathfrak{m})$ .

Furthermore, if  $a \equiv 1 \pmod{m}$  then clearly  $\Phi_{\mathfrak{m}}(a\mathbb{Z}) = 1 \in (\mathbb{Z}/m\mathbb{Z})^*$ . Since  $P_{\mathbb{Q},1}(\mathfrak{m})$  is generated by ideals of the form  $a\mathbb{Z}$  with  $a \equiv 1 \pmod{m}$  this shows that

$$\ker(\Phi_{\mathfrak{m}}) = P_{\mathbb{Q},1}(\mathfrak{m}).$$

In particular,  $\mathbb{Q}(\zeta_m)$  is the ray class field of  $m\infty$ .

(ii) As a next step we can determine the conductor of  $\mathbb{Q} \subset \mathbb{Q}(\zeta_m)$ . We will see that

$$f(\mathbb{Q}(\zeta_m)/\mathbb{Q}) = \begin{cases} 1 & \text{if } m \leq 2 \\ (m/2)\infty & \text{if } m = 2n, n > 1 \text{ odd} \\ m\infty & \text{otherwise} \end{cases}$$

where  $\infty$  is the embedding of  $\mathbb{Q}$  into  $\mathbb{C}$ .

The case  $m \leq 2$  is trivial. Suppose  $m > 2$ , write  $f\infty$  for the conductor of  $\mathbb{Q}(\zeta_m)/\mathbb{Q}$ . Since we already know that  $m$  is a suitable modulus we have that  $f \mid m$ . Now  $f\infty$  is a suitable modulus for both  $\mathbb{Q}(\zeta_m)$  and  $\mathbb{Q}(\zeta_f)$  as field extensions of  $\mathbb{Q}$ . By part (i)  $\ker(\Phi_{\mathbb{Q}(\zeta_f)/\mathbb{Q}}(f\infty)) = P_{\mathbb{Q},1}(f\infty)$  and thus we have

$$P_{\mathbb{Q},1}(f\infty) = \ker(\Phi_{\mathbb{Q}(\zeta_f)/\mathbb{Q},f\infty}) \subset \ker(\Phi_{\mathbb{Q}(\zeta_m)/\mathbb{Q},f\infty}).$$

By Corollary 4.12 we have  $\mathbb{Q}(\zeta_m) \subset \mathbb{Q}(\zeta_f)$  and  $f \mid m$  implies that  $\mathbb{Q}(\zeta_m) = \mathbb{Q}(\zeta_f)$ . For  $\phi$  the Euler-function we get  $\phi(m) = \phi(f)$ , since these are the orders of the respective Galois groups over  $\mathbb{Q}$ . The smallest possibilities for  $f$  such that  $\phi(m) = \phi(f)$  holds are

$$f = \begin{cases} (m/2) & \text{if } m = 2n, n > 1 \text{ odd,} \\ m & \text{otherwise.} \end{cases}$$

(We can see this using the formula  $\phi(mf) = \phi(m)\phi(f)\frac{d}{\phi(d)}$ , where  $d = \gcd(m, f)$ .) Since for this choice of  $f$ ,  $\mathbb{Q}(\zeta_f) = \mathbb{Q}(\zeta_m)$ , by part (i) we deduce that  $f\infty$  is a fitting modulus for  $\mathbb{Q} \subset \mathbb{Q}(\zeta_m)$  and thus is the conductor.

We conclude this section by proving some classical results.

**Theorem 4.15** (Kronecker-Weber, [Cox89], Theorem 8.8). For every abelian extension  $\mathbb{Q} \subset L$ , there exists a positive integer  $m$  such that  $L \subset \mathbb{Q}(\zeta_m)$ , where  $\zeta_m = e^{\frac{2\pi i}{m}}$ .

*Proof.* By Artin reciprocity there is some modulus  $\mathfrak{m}$  in  $\mathbb{Q}$  such that  $P_{\mathbb{Q},1}(\mathfrak{m}) \subset \Phi_{L/\mathbb{Q},\mathfrak{m}}$ , where  $\Phi_{L/\mathbb{Q},\mathfrak{m}}$  is the Artin map. We may assume that  $\mathfrak{m}$  takes the form  $m\infty$  for some  $m > 0$ , where  $\infty$  is once again the embedding  $\mathbb{Q} \hookrightarrow \mathbb{R}$ . But now we have

$$P_{\mathbb{Q},1}(\mathfrak{m}) = \Phi_{\mathbb{Q}(\zeta_m)/\mathbb{Q},\mathfrak{m}} \subset \Phi_{L/\mathbb{Q},\mathfrak{m}}$$

since  $\mathbb{Q}(\zeta_m)$  is the ray class field corresponding to  $m\infty$ . By Corollary 4.12, we deduce  $L \subset \mathbb{Q}(\zeta_m)$ .  $\square$

**Remark 4.16.** Finding a suitable modulus for a given field extension  $\mathbb{Q} \subset L$  can be quite hard, so sometimes it is easier to find a cyclotomic field extension  $\mathbb{Q}(\zeta_m)$  containing  $L$ . We then know that  $m\infty$  is also a suitable modulus for  $L$ . This follows by Proposition 4.13, since if  $\ker(\Phi_{\mathbb{Q}(\zeta_m)/\mathbb{Q}, m\infty})$  is a congruence subgroup, then so is  $\ker(\Phi_{L/\mathbb{Q}, m\infty})$ .

**Theorem 4.17** (Existence of the Hilbert class field, [Cox89], Theorem 8.10). Let  $K$  be a number field, then there exists a maximal unramified abelian extension  $K \subset L$ . We call  $L$  the *Hilbert class field* of  $K$ .

*Proof.* First notice that for any unramified abelian extension the Conductor Theorem 4.8 tells us that the kernel of the Artin map is a congruence subgroup for  $\mathfrak{m} = 1$ . Applying the Existence Theorem 4.10 to the modulus  $\mathfrak{m} = 1$  and the congruence subgroup  $P_K$  gives an unramified abelian extension  $K \subset L$ . If  $L'$  is another unramified abelian extension then

$$P_K = \ker(\Phi_{L/\mathbb{Q}, 1}) \subset \ker(\Phi_{L'/\mathbb{Q}, 1})$$

so by Corollary 4.12 we have  $L' \subset L$ .  $\square$

**Example 4.18.** (i) If  $K$  is a number field with class number 1, i.e.  $O_K$  a UFD, then the Hilbert class field is the field itself. This is easy to see, since Artin reciprocity gives us an isomorphism  $P_K/I_K \xrightarrow{\sim} \text{Gal}(L/K)$  if  $L$  is the Hilbert class field.

(ii) Let  $K = \mathbb{Q}(\sqrt{-5})$ , then the class number is 2. The Hilbert class field is therefore some quadratic extension of  $K$ . If we write  $L = K(\sqrt{m})$  for the Hilbert class field, then some elementary considerations about ramification tell us that  $m$  should divide  $-5$  so the options are  $-1, 5$  which both result in the same field extension. By existence of the Hilbert class field, we know that there is an unramified degree 2 extension of  $\mathbb{Q}(\sqrt{-5})$ , which in this case must be  $\mathbb{Q}(\sqrt{-5}, i)$ .

One can check that the field extension is indeed unramified by considering the extension  $\mathbb{Q} \subset \mathbb{Q}(\sqrt{-5}, i)$ . We have three intermediate field extensions and the only primes that ramify in those are 2 and 5. Using some elementary considerations about ramification degrees and Dedekind's criterion, we can deduce  $\mathbb{Q}(\sqrt{-5}) \subset \mathbb{Q}(\sqrt{-5}, i)$  is indeed unramified.

## 5 The Legendre Symbol

Let  $K$  be a number field containing a primitive  $n$ -th root of unity  $\zeta_n$ . In this section we want to define the  $n$ -th power Legendre symbol and it will be analogous to the definition of the cubic Legendre symbol in Section 2.

Let  $\mathfrak{p}$  be a prime ideal in  $O_K$ . Suppose  $n$  is coprime to  $\mathfrak{p}$ , i.e. the ideal generated by  $n$  is coprime to  $\mathfrak{p}$ . Then we can deduce that  $1, \zeta_n, \dots, \zeta_n^{n-1}$  are distinct modulo  $\mathfrak{p}$  since  $X^n - 1$  is

separable modulo  $\mathfrak{p}$ , i.e.,  $X^n - 1$  and  $nX^{n-1}$  do not have any common zeros modulo  $\mathfrak{p}$ . This shows that  $\{1, \zeta_n, \dots, \zeta_n^{n-1}\}$  is a  $n$  element subgroup of  $(O_K/\mathfrak{p})^*$  and thus  $n \mid N(\mathfrak{p}) - 1$ , since  $N(\mathfrak{p}) - 1 = |(O_K/\mathfrak{p})^*|$ . We will frequently denote the subgroup  $\{1, \zeta_n, \dots, \zeta_n^{n-1}\}$  by  $\mu_n$ . For an element  $\alpha \in O_K$  which is coprime to  $\mathfrak{p}$ , Fermat's Little Theorem tells us that

$$\alpha^{N(\mathfrak{p})-1} \equiv 1 \pmod{\mathfrak{p}}.$$

Now we know that  $x = \alpha^{(N(\mathfrak{p})-1)/n}$  is a  $n$ -th root of unity modulo  $\mathfrak{p}$ , i.e.,

$$\alpha^{(N(\mathfrak{p})-1)/n} \equiv 1, \zeta_n, \dots, \zeta_n^{n-1} \pmod{\mathfrak{p}}.$$

**Definition 5.1.** We define the  $n$ -th power Legendre symbol  $\left(\frac{\alpha}{\mathfrak{p}}\right)_n$  to be the root of unity satisfying

$$\alpha^{(N(\mathfrak{p})-1)/n} \equiv \left(\frac{\alpha}{\mathfrak{p}}\right)_n \pmod{\mathfrak{p}}.$$

We can extend this map by multiplicativity. Let  $\mathfrak{m}$  be a modulus in  $K$  such that  $n\alpha$  divides  $\mathfrak{m}$ . Then we have a homomorphism

$$\left(\frac{\alpha}{\cdot}\right)_n : I_K(\mathfrak{m}) \longrightarrow \mu_n$$

$$\left(\frac{\alpha}{\mathfrak{a}}\right)_n = \prod_{i=1}^r \left(\frac{\alpha}{\mathfrak{p}_i}\right)_n^{\nu_i}$$

for  $\mathfrak{a} = \mathfrak{p}_1^{\nu_1} \dots \mathfrak{p}_r^{\nu_r}$ .

We now want to study the relation of the  $n$ -th power Legendre symbol to the Artin map. Let  $K$  be a number field containing a primitive  $n$ -th root of unity, fix  $\sqrt[n]{\alpha}$  as some  $n$ -th root of  $\alpha$ , and let  $L = K(\sqrt[n]{\alpha})$ . For an unramified prime ideal  $\mathfrak{p}$  coprime to  $n\alpha$  we get

$$\begin{aligned} \left(\frac{L/K}{\mathfrak{p}}\right)(\sqrt[n]{\alpha}) &\equiv (\sqrt[n]{\alpha})^{N(\mathfrak{p})} \pmod{\mathfrak{P}} \\ &\equiv \alpha^{(N(\mathfrak{p})-1)/n} \sqrt[n]{\alpha} \pmod{\mathfrak{P}} \end{aligned}$$

where  $\mathfrak{P}$  is some prime ideal lying over  $\mathfrak{p}$ .

Thus we can deduce that

$$\left(\frac{L/K}{\mathfrak{p}}\right)(\sqrt[n]{\alpha}) \equiv \left(\frac{\alpha}{\mathfrak{p}}\right)_n \sqrt[n]{\alpha} \pmod{\mathfrak{P}}$$

But since  $\left(\frac{L/K}{\mathfrak{p}}\right)(\sqrt[n]{\alpha})$  and  $\left(\frac{\alpha}{\mathfrak{p}}\right)_n \sqrt[n]{\alpha}$  must be  $n$ -th roots of  $\alpha$  and these are distinct modulo  $\mathfrak{P}$ , we can deduce that

$$\left(\frac{L/K}{\mathfrak{p}}\right)(\sqrt[n]{\alpha}) = \left(\frac{\alpha}{\mathfrak{p}}\right)_n \sqrt[n]{\alpha}. \quad (3)$$

**Theorem 5.2.** [Weak Reciprocity, [Cox89], Theorem 8.11] Let  $K$  be a number field containing a primitive  $n$ -th root of unity  $\zeta_n^k$ , and let  $L = K(\sqrt[n]{\alpha})$ , where  $\alpha \in O_K$  is nonzero and  $\sqrt[n]{\alpha}$  is a  $n$ -th root of unity. Assume that  $\mathfrak{m}$  is a modulus divisible by all primes of  $K$  containing  $n\alpha$  and that  $\ker(\Phi_{\mathfrak{m}})$  is a congruence subgroup for  $\mathfrak{m}$ . Then there is a commutative diagram

$$\begin{array}{ccc} I_K(\mathfrak{m}) & \xrightarrow{\Phi_{\mathfrak{m}}} & \text{Gal}(L/K) \\ & \searrow (\frac{\alpha}{\cdot})_n & \downarrow \\ & & \mu_n \end{array}$$

where  $\text{Gal}(L/K) \rightarrow \mu_n$  is the identification via  $\sigma \mapsto \zeta_n^k$ , such that  $\sigma(\sqrt[n]{\alpha}) = \zeta_n^k \sqrt[n]{\alpha}$ . Thus, if  $G$  is the image of  $\text{Gal}(L/K)$  in  $\mu_n$ , then the  $n$ -th power Legendre symbol  $(\frac{\alpha}{\cdot})_n$  induces a surjective homomorphism

$$\left(\frac{\alpha}{\cdot}\right)_n : I_K(\mathfrak{m})/P_{K,1}(\mathfrak{m}) \longrightarrow G \subset \mu_n.$$

*Proof.* This follows from

$$\left(\frac{L/K}{\mathfrak{p}}\right)(\sqrt[n]{\alpha}) = \left(\frac{\alpha}{\mathfrak{p}}\right)_n \sqrt[n]{\alpha}.$$

for all  $\mathfrak{p} \in I_K(\mathfrak{m})$ . □

A much stronger tool is the strong reciprocity law or also called the general reciprocity law for the  $n$ -th power Legendre symbol. To state this theorem we first need to define the  $n$ -th Hilbert Symbol. For a local field  $K$  with maximal ideal  $\mathfrak{p}$ , we have

$$\left(\frac{\cdot, \cdot}{\mathfrak{p}}\right)_n : K^*/K^{*n} \times K^*/K^{*n} \longrightarrow \mu_n.$$

We will use the completion  $K_{\mathfrak{p}}$  of  $K$  at the prime ideal  $\mathfrak{p}$  as our local field. The Hilbert symbol is a tool in local class field theory and its definition requires some knowledge about local class field theory and group cohomology. Instead of defining it here, we refer to the Appendix B, where the Hilbert symbol is properly defined.

**Theorem 5.3.** [Strong Reciprocity, [Cox89], Theorem 8.14] Let  $K$  be a number field containing a primitive  $n$ -th root of unity  $\zeta_n^k$  and let  $\alpha, \beta \in O_K$  be relatively prime to each other and to  $n$ . Then

$$\left(\frac{\alpha}{\beta}\right)_n \cdot \left(\frac{\beta}{\alpha}\right)_n^{-1} = \prod_{\mathfrak{p}|n\infty} \left(\frac{\alpha, \beta}{\mathfrak{p}}\right)_n$$

where  $\infty$  is the product of the real infinite primes of  $K$ .

*Proof.* See Appendix C. □

We will see an application of the strong reciprocity law in Section 7.

## 6 Quadratic Reciprocity (revisited)

After getting to know the main theorems of global class field theory we want to deduce the classical result of quadratic reciprocity from the weak reciprocity law.

Let us first remind ourselves of the definition of the quadratic Legendre symbol.

**Definition 6.1.** Let  $p$  be an odd prime, and let  $a \in \mathbb{Z}$ . We define the Legendre symbol to be

$$\left(\frac{a}{p}\right) = \begin{cases} 1 & \text{if } p \nmid a \text{ and } a \text{ is a quadratic residue mod } p \\ -1 & \text{if } p \nmid a \text{ and } a \text{ is not a quadratic residue mod } p \\ 0 & \text{if } p \mid a. \end{cases}$$

**Remark 6.2.** Notice that for  $K = \mathbb{Q}$  we have that  $N(p) = p$  for all primes  $p \in \mathbb{Q}$ . Thus by Lemma 1.4, the definition of the  $n$ -th power Legendre symbol is consistent with our previous definition of the  $n$ -th power Legendre symbol for  $n = 2$ .

To avoid confusion, we will always write  $\left(\frac{\cdot}{p}\right)_2$  for the quadratic Legendre symbol from now on.

**Lemma 6.3.** Let  $p$  be an odd prime and  $\mathbb{Q}(\sqrt{m})$ ,  $m \in \mathbb{Z}$  squarefree, a quadratic field extension of  $\mathbb{Q}$  with  $p$  being the only finite prime that possibly ramifies in  $\mathbb{Q}(\sqrt{m})$ . Then  $m = (-1)^{\frac{p-1}{2}} p$ .

*Proof.* This follows from the fact that a prime of  $\mathbb{Z}$  ramifies in  $\mathbb{Q}(\sqrt{m})$  if and only if it divides the discriminant. Hence the discriminant is only divisible by  $p$  and is thus  $\pm p$ . If  $m \not\equiv 1 \pmod{4}$  then the discriminant would be  $4m$  and not  $\pm p$ . Thus we deduce that  $m \equiv 1 \pmod{4}$  and the discriminant is  $m$ , i.e.,  $m = \pm p$ . But now if  $p = m$ , then  $p \equiv 1 \pmod{4}$  and  $m = (-1)^{\frac{p-1}{2}} p$  holds, and if  $p = -m$  then  $p \equiv 3 \pmod{4}$  and  $m = (-1)^{\frac{p-1}{2}} p$  also holds.  $\square$

**Lemma 6.4.** Let  $m > 0$  be an integer and  $\infty$  be the embedding of  $\mathbb{Q}$  into  $\mathbb{R}$ . Then the map  $a \mapsto a\mathbb{Z}$  for  $a > 0$  induces a surjective homomorphism  $(\mathbb{Z}/m\mathbb{Z})^* \rightarrow I_{\mathbb{Q}}(m\infty)/P_{\mathbb{Q},1}(m\infty)$ .

*Proof.* We first show that the map is well-defined. If  $a, b \in (\mathbb{Z}/m\mathbb{Z})^*$ ,  $a \equiv b \pmod{m}$ ,  $a, b > 0$  then there is an integer  $c > 0$ , such that  $ac \equiv bc \equiv 1 \pmod{m}$ . Now  $b\mathbb{Z} \cdot ac\mathbb{Z} = a\mathbb{Z} \cdot bc\mathbb{Z}$ . Since  $ac\mathbb{Z}$  and  $bc\mathbb{Z}$  are in  $P_{\mathbb{Q},1}(m\infty)$  we have  $[a\mathbb{Z}] = [b\mathbb{Z}]$  in  $I_{\mathbb{Q}}(m\infty)/P_{\mathbb{Q},1}(m\infty)$ .

Notice that the map is clearly a homomorphism. For surjectivity, first notice that  $\mathbb{Z}$  is a PID, thus all fractional ideals in  $I_{\mathbb{Q}}(m\infty)$  take the form  $\frac{a}{b}\mathbb{Z}$ , with  $a, b > 0$  coprime to  $m$ . Let  $\frac{a}{b}\mathbb{Z} \in I_{\mathbb{Q}}(m\infty)$ , with  $a, b > 0$  coprime to  $m$ , written in lowest terms. Then there is an integer  $c > 0$ , such that  $cb \equiv 1 \pmod{m}$ . Therefore we have  $[\frac{a}{b}\mathbb{Z}] = [\frac{a}{b}\mathbb{Z} \cdot bc\mathbb{Z}] = [ac\mathbb{Z}]$ ,  $ac > 0$  coprime to  $m$ , i.e.,  $ac \in (\mathbb{Z}/m\mathbb{Z})^*$  is being mapped to  $[\frac{a}{b}\mathbb{Z}]$  and the map is surjective.  $\square$

**Theorem 6.5** (Quadratic reciprocity). Let  $p, q$  be distinct odd primes. Then

$$\left(\frac{p}{q}\right)_2 = \left(\frac{q}{p}\right)_2 (-1)^{\frac{p-1}{2} \cdot \frac{q-1}{2}}$$

*Proof.* We want to prove the equivalent statement

$$\left(\frac{p^*}{q}\right)_2 = \left(\frac{q}{p}\right)_2$$

for  $p^* = (-1)^{\frac{p-1}{2}} p$ . The equivalence follows from  $\left(\frac{-1}{q}\right)_2 = (-1)^{\frac{q-1}{2}}$ .

Let  $\infty$  be the embedding of  $\mathbb{Q}$  into  $\mathbb{R}$ . First we want to show that  $p\infty$  is a modulus for the quadratic field extension  $\mathbb{Q} \subset \mathbb{Q}(\sqrt{p^*})$ , i.e.,  $p\infty$  is divisible by all primes that possibly ramify and the kernel of the Artin map is a congruence subgroup.

To this end, we show it is the unique quadratic field extension of  $\mathbb{Q}$  that lies in the cyclotomic field  $\mathbb{Q}(\zeta_p)$ , where  $\zeta_p$  is a primitive  $p$ -th root of unity. Since  $\text{Gal}(\mathbb{Q}(\zeta_p)/\mathbb{Q})$  is cyclic of order  $p-1$ , there is a unique quadratic field extension of  $\mathbb{Q}$  that lies in  $\mathbb{Q}(\zeta_p)$ . Recall that any quadratic field extension of  $\mathbb{Q}$  can be written as  $\mathbb{Q}(\sqrt{m})$ , with  $m$  squarefree. Since  $p\infty$  is the conductor for  $\mathbb{Q} \subset \mathbb{Q}(\zeta_p)$  it is also a fitting modulus for  $\mathbb{Q} \subset \mathbb{Q}(\sqrt{m})$ . This means the only finite prime of  $\mathbb{Q}$  that possibly ramifies in  $\mathbb{Q}(\zeta_p)$  is  $p$ . By Lemma 6.3 we have that  $m = p^*$ .

By the weak reciprocity law, Theorem 5.2,  $\left(\frac{p^*}{\cdot}\right)_2$  defines a surjective homomorphism  $I_{\mathbb{Q}}(p\infty)/P_{\mathbb{Q},1}(p\infty) \rightarrow \{\pm 1\}$ , so composing with the map from Lemma 6.4 yields a surjective homomorphism  $(\mathbb{Z}/p\mathbb{Z})^* \rightarrow \{\pm 1\}$ . But the Legendre symbol  $\left(\frac{\cdot}{p}\right)_2$  is also a surjective homomorphism  $(\mathbb{Z}/p\mathbb{Z})^* \rightarrow \{\pm 1\}$ . Thus they must be the same and in particular,

$$\left(\frac{p^*}{q}\right)_2 = \left(\frac{q}{p}\right)_2. \quad \square$$

We can also derive the supplementary law for quadratic reciprocity from the weak reciprocity law.

**Proposition 6.6.** Let  $p$  be an odd prime. Then

$$\left(\frac{2}{p}\right)_2 = (-1)^{\frac{p^2-1}{8}}.$$

*Proof.* First notice that

$$\begin{aligned} (\mathbb{Z}/8\mathbb{Z})^* &\longrightarrow \{\pm 1\} \\ a &\mapsto (-1)^{\frac{a^2-1}{8}} \end{aligned}$$

defines a surjective homomorphism with kernel  $\{\pm 1\} \subset (\mathbb{Z}/8\mathbb{Z})^*$ . The map is well-defined since for all  $a \in (\mathbb{Z}/8\mathbb{Z})^*$  we have  $a^2 \equiv 1 \pmod{8}$ . Since there are only 4 elements we can check explicitly that the map defines a group homomorphism.

Our goal is to show that  $\left(\frac{2}{\cdot}\right)_2$  also defines such a homomorphism when precomposed with  $\varphi : (\mathbb{Z}/8\mathbb{Z})^* \longrightarrow I_{\mathbb{Q}}(8\infty)/P_{\mathbb{Q}}(8\infty)$ ,  $a \mapsto [a\mathbb{Z}]$ , as seen in Lemma 6.4. In order to use weak reciprocity we need to show that  $\mathfrak{m} = 8\infty$  in  $\mathbb{Q}$  is a suitable modulus for the field extension  $\mathbb{Q} \subset \mathbb{Q}(\sqrt{2})$ , where  $\infty$  is the embedding  $\mathbb{Q} \hookrightarrow \mathbb{R}$ . Furthermore, we want to show that the kernel of the Artin map of  $\mathbb{Q} \subset \mathbb{Q}(\sqrt{2})$  with modulus  $8\infty$  corresponds to  $\{\pm 1\} \subset (\mathbb{Z}/8\mathbb{Z})^*$ .

Let  $\pi : I_{\mathbb{Q}}(8\infty) \rightarrow I_{\mathbb{Q}}(8\infty)/P_{\mathbb{Q}}(8\infty)$  be the projection map. Now if  $H'$  is the image of the subgroup  $\{\pm 1\} \subset (\mathbb{Z}/8\mathbb{Z})^*$  in  $I_K(8\infty)/P_K(8\infty)$  then set  $H = \pi^{-1}(H')$ . Let  $L$  be the unique field extension of  $K$  given by the Existence Theorem 4.10 with regard to  $H$ . We want to show that  $L = \mathbb{Q}(\sqrt{2})$ .

We can see that  $L \subsetneq \mathbb{Q}(\zeta_8)$  since  $\mathbb{Q}(\zeta_8)$  is the ray class field for  $\mathfrak{m} = 8\infty$  and  $H$  is nontrivial. Using that  $\text{Gal}(\mathbb{Q}(\zeta_8)/\mathbb{Q}) \cong C_2 \times C_2$  we are left with the options  $\mathbb{Q}(i)$ ,  $\mathbb{Q}(\sqrt{2})$ , and  $\mathbb{Q}(\sqrt{-2})$ .

We have  $\varphi(-1) = \varphi(7) = [7\mathbb{Z}] \in H'$ , so by construction  $7\mathbb{Z} \in H$  is in the kernel of the Artin map  $\Phi_{L/\mathbb{Q}, 8\infty}$ . We check explicitly whether this can be the case for  $L$  equal  $\mathbb{Q}(i)$ ,  $\mathbb{Q}(\sqrt{2})$ , or  $\mathbb{Q}(\sqrt{-2})$ .

Note that 7 is inert in  $\mathbb{Q} \subset \mathbb{Q}(i)$ ,  $\mathbb{Q}(\sqrt{-2})$ . The Artin symbols of the field extensions  $\mathbb{Q} \subset \mathbb{Q}(i)$ ,  $\mathbb{Q}(\sqrt{-2})$  are uniquely determined by

$$\begin{aligned} \left(\frac{\mathbb{Q}(i)/\mathbb{Q}}{7}\right)(i) &\equiv i^7 \equiv -i \pmod{7}, \\ \left(\frac{\mathbb{Q}(\sqrt{2})/\mathbb{Q}}{7}\right)(\sqrt{-2}) &\equiv (\sqrt{-2})^7 \equiv -\sqrt{-2} \pmod{7}. \end{aligned}$$

However, in both cases the map induced by the complex conjugation satisfies the same relation. We deduce that 7 is not in the kernel of the corresponding Artin maps. Therefore we have that  $L = \mathbb{Q}(\sqrt{2})$  and thus  $8\infty$  is a suitable modulus for  $\mathbb{Q}(\sqrt{2})$ .

Finally, by weak reciprocity  $\left(\frac{2}{\cdot}\right)_2$  defines a surjective homomorphism  $I_{\mathbb{Q}}(8\infty)/P_{\mathbb{Q},1}(8\infty) \rightarrow \text{Gal}(\mathbb{Q}(\sqrt{2})/\mathbb{Q}) \cong \{\pm 1\}$ . If we precompose this map with  $(\mathbb{Z}/8\mathbb{Z})^* \rightarrow I_{\mathbb{Q}}(8\infty)/P_{\mathbb{Q}}(8\infty)$  we get a surjective homomorphism

$$(\mathbb{Z}/8\mathbb{Z})^* \rightarrow \{\pm 1\}$$

with kernel  $\{\pm 1\} \subset (\mathbb{Z}/8\mathbb{Z})^*$ , by our construction.

But since

$$\begin{aligned} (\mathbb{Z}/8\mathbb{Z})^* &\rightarrow \{\pm 1\} \\ a &\mapsto (-1)^{\frac{a^2-1}{8}} \end{aligned}$$

also defines a surjective homomorphism with kernel  $\{\pm 1\} \subset (\mathbb{Z}/8\mathbb{Z})^*$ , we deduce that they are the same map. In particular for  $p$  an odd prime

$$\left(\frac{2}{p}\right)_2 = (-1)^{\frac{p^2-1}{8}}. \quad \square$$

## 7 Cubic Reciprocity (revisited)

In this section we want to derive cubic reciprocity from the theorems of class field theory. We will give two proofs, the first one using the Hilbert symbol (see Appendix B) and the second only using the weak reciprocity law.

Once again, we find ourselves in the setting where  $K = \mathbb{Q}(\omega)$  where  $\omega = e^{\frac{2\pi i}{3}}$  and  $O_K = \mathbb{Z}[\omega]$ . The definition of the cubic Legendre symbol in Section 2 is consistent with the definition of the  $n$ -th power Legendre symbol in Section 5. Recall that for  $\pi$  a prime in  $\mathbb{Z}[\omega]$  and  $\alpha \in \mathbb{Z}[\omega]$ ,

$$\alpha^{\frac{N(\pi)-1}{3}} \equiv \left(\frac{\alpha}{\pi}\right)_3 \pmod{\pi}$$

Throughout this section, instead of assuming  $\pi, \theta$  are primary ( $-1 \pmod{3}$ ), we will assume  $\pi, \theta \equiv 1 \pmod{3}$ . This will not change the statement by Remark 2.12.

**Theorem 7.1** (Cubic reciprocity). Let  $\pi, \theta$  be primes in  $O_K$ , coprime to each other, such that  $\pi, \theta \equiv 1 \pmod{3}$  in  $\mathbb{Z}[\omega]$ . Then

$$\left(\frac{\pi}{\theta}\right)_3 = \left(\frac{\theta}{\pi}\right)_3.$$

### Proof 1

This proof will use the strong reciprocity law and the properties of the Hilbert symbol. The reader is encouraged to take a look at the Appendix first. Remember that  $3 = -\omega^2(1 - \omega)^2$ .

We state two properties of the Hilbert symbol that we will use here, they will come up again later in the Appendix B.

**Proposition 7.2.** Let  $K$  be a local field and let  $\alpha, \alpha', \beta, \beta' \in K^*$ .

- (i) Bi-multiplicativity:  $\left(\frac{\alpha\alpha', \beta}{\mathfrak{p}}\right)_n = \left(\frac{\alpha, \beta}{\mathfrak{p}}\right)_n \cdot \left(\frac{\alpha', \beta}{\mathfrak{p}}\right)_n$  and  $\left(\frac{\alpha, \beta\beta'}{\mathfrak{p}}\right)_n = \left(\frac{\alpha, \beta}{\mathfrak{p}}\right)_n \cdot \left(\frac{\alpha, \beta'}{\mathfrak{p}}\right)_n$
- (ii)  $\left(\frac{\alpha, -\alpha}{\mathfrak{p}}\right)_n = 1$  and  $\left(\frac{\alpha, 1-\alpha}{\mathfrak{p}}\right)_n = 1$

**Lemma 7.3.** Let  $\lambda = 1 - \omega$ , let  $K_\lambda$  denote the completion of  $K$  at  $\lambda$  and  $O_\lambda$  the valuation ring, i.e., the ring of integers. For  $\alpha, \alpha' \in O_\lambda^*$  and any  $\beta \in K_\lambda^*$ , we have

$$\alpha \equiv \alpha' \pmod{\lambda^4 O_\lambda} \implies \left(\frac{\alpha, \beta}{\lambda}\right)_3 = \left(\frac{\alpha', \beta}{\lambda}\right)_3.$$

*Proof.* We begin by showing that if  $\gamma \equiv 1 \pmod{\lambda^4 O_\lambda}$  then  $\gamma$  is a cube in  $O_\lambda$ . Clearly, if  $\gamma \equiv 1 \pmod{\lambda^4 O_\lambda}$ , then  $\gamma$  is a cube modulo  $\lambda^3$ .

We can lift the cube roots of  $\gamma$  modulo  $\lambda^n$  to cube roots of  $\gamma$  modulo  $\lambda^{n+1}$  for  $n \geq 4$ . To do this, let  $u_n \in O_\lambda$  such that  $\gamma = u_n^3 \pmod{\lambda^n O_\lambda}$ , i.e.,  $u_n^3 = \gamma + b\lambda^n$  for some  $b \in O_\lambda$ . For any  $a \in O_\lambda$

$$\begin{aligned} (u_n + a\lambda^{n-2})^3 &= u_n^3 + 3a\lambda^{n-2}u_n^2 + 3a^2\lambda^{(n-2)2}u_n + a^3\lambda^{(n-2)3} \equiv u_n^3 + 3a\lambda^{n-2}u_n^2 \pmod{\lambda^{n+1}} \\ &\equiv \gamma + b\lambda^n - \omega^2\lambda^2 a\lambda^{n-2}u_n^2 \pmod{\lambda^{n+1}} \end{aligned}$$

where we used  $3 = -\omega^2(1 - \omega)^2$  and  $(n - 2)3 \geq n + 1$  for  $n \geq 4$ . We can choose  $a \in O_\lambda$  such that  $b - au_n^2\omega^2 \equiv 0 \pmod{\lambda}$ , since  $u_n^2\omega^2 \not\equiv 0 \pmod{\lambda}$ .

We get that  $u_{n+1} := u_n + a\lambda^{n-2}$  satisfies

- (i)  $u_{n+1}^3 \equiv \gamma \pmod{\lambda^{n+1}O_\lambda}$
- (ii)  $u_{n+1} \equiv u_n \pmod{\lambda^{n-2}O_\lambda}$ .

By definition of the completion at  $\lambda$  as an inverse limit we can find a cube root of  $\gamma$  in  $O_\lambda$ .

Let  $u \in O_\lambda$  be a cube root of  $\gamma$ , then we have

$$\left(\frac{\gamma, \beta}{\lambda}\right)_3 = \left(\frac{u, \beta}{\lambda}\right)_3^3 = 1$$

for all  $\beta \in K_\lambda^*$ .

Now let  $\alpha, \alpha^* \in O_\lambda^*$  and  $\alpha \equiv \alpha' \pmod{\lambda^4 O_\lambda}$  then  $\alpha^{-1}\alpha' \equiv 1 \pmod{\lambda^4 O_\lambda}$  and

$$\left(\frac{\alpha^{-1}\alpha', \beta}{\lambda}\right)_3 = 1.$$

Hence

$$\left(\frac{\alpha, \beta}{\lambda}\right)_3 = \left(\frac{\alpha', \beta}{\lambda}\right)_3$$

for all  $\beta \in K_\lambda^*$ . □

*Proof of Cubic reciprocity.* First notice that the strong reciprocity law (Theorem 5.3) directly gives us

$$\left(\frac{\pi}{\theta}\right)_3 \cdot \left(\frac{\theta}{\pi}\right)_3^{-1} = \left(\frac{\pi, \theta}{\lambda}\right)_3.$$

with  $\lambda = 1 - \omega$ , since  $\lambda$  is the only prime dividing 3 in  $\mathbb{Z}[\omega]$  and there are no real infinite primes of  $K$ . Since 3 and  $\lambda^2$  are associate, we have  $\pi, \theta \equiv 1 \pmod{\lambda^2}$ . We will proceed to show

$$\pi, \theta \equiv 1 \pmod{\lambda^2} \implies \left(\frac{\pi, \theta}{\lambda}\right)_3 = 1.$$

Let  $K_\lambda$  denote the completion of  $K$  at  $\lambda$ . If we write  $\pi = 1 + a\lambda^2$ , then

$$\pi \equiv 1 + a\theta\lambda^2 \pmod{\lambda^4 O_\lambda}. \tag{4}$$

By Lemma 7.2 (iii) we get

$$\left(\frac{1 + a\theta\lambda^2, -a\theta\lambda^2}{\lambda}\right)_3 = 1 \text{ and } \left(\frac{\pi, -a\lambda^2}{\lambda}\right)_3 = 1$$

Using Lemma 7.3 and (4) we deduce that

$$\left(\frac{\pi, -a\theta\lambda^2}{\lambda}\right)_3 = \left(\frac{1 + a\theta\lambda^2, -a\theta\lambda^2}{\lambda}\right)_3 = 1.$$

Finally, we use the multiplicativity of the Hilbert symbol to get

$$1 = \left(\frac{\pi, -a\theta\lambda^2}{\lambda}\right)_3 = \left(\frac{\pi, -a\lambda^2}{\lambda}\right)_3 \cdot \left(\frac{\pi, \theta}{\lambda}\right)_3 = \left(\frac{\pi, \theta}{\lambda}\right)_3$$

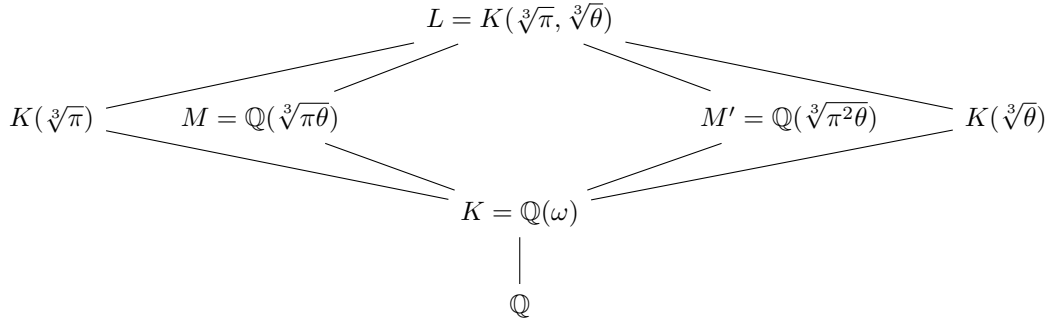
which proves the cubic reciprocity law. □

## Proof 2

We now illustrate a second proof that does not need the definition of the Hilbert symbol.

The vague idea is to proceed as in Section 6. For a prime  $\pi$ , we can view  $(\frac{\cdot}{\pi})_3$  as a homomorphism from  $O_K/\langle\pi\rangle$  to  $\{1, \omega, \omega^2\}$  and  $(\frac{\pi}{\cdot})_3$  as homomorphism from  $I_K(\mathfrak{m})/P_{K,1}(\mathfrak{m})$  to  $\{1, \omega, \omega^2\}$  for some fitting modulus  $\mathfrak{m}$ , using the weak reciprocity law for some cubic extension of  $K$ . The problem is that we clearly always have (none or) at least two different homomorphisms from any group into  $\{1, \omega, \omega^2\}$  since it has a nontrivial automorphism. Instead, we want to compare homomorphisms induced by  $(\frac{\cdot}{\pi})_3, (\frac{\cdot}{\theta})_3, (\frac{\pi}{\cdot})_3, (\frac{\theta}{\cdot})_3$  for primes  $\pi, \theta$ . We will proceed to do this by looking at suitable field extensions of  $K$ .

Let  $K = \mathbb{Q}(\omega)$ ,  $\omega = e^{\frac{2\pi i}{3}}$  and  $\pi, \theta$  primes of unequal norm in  $K$  satisfying  $\pi, \theta \equiv 1 \pmod{3}$ . We will not specify which third root we mean when writing  $K(\sqrt[3]{\pi})$  or  $K(\sqrt[3]{\theta})$  since they will result in the same field extension ( $K$  contains a third root of unity). In order to study the cubic Legendre symbols we consider the following field extensions:



First notice that  $[L : K] = 9$  and  $\text{Gal}(L/K) \cong C_3 \times C_3$  so indeed we have 4 intermediate fields of degree 3. The key is that either  $M \subset L$  or  $M' \subset L$  is unramified. To check this, the only primes we need to consider are the primes over 3,  $\pi$  and  $\theta$ . Since the proof is rather technical, we will state this result here and prove it at the end of the section.

**Lemma 7.4.** Let  $K, L, M, M'$  be as above. Let  $\pi, \theta$  be primes in  $O_K$  of distinct norm satisfying  $\pi, \theta \equiv 1 \pmod{3}$ . Then either  $M \subset L$  or  $M' \subset L$  is unramified.

We need one more simple lemma to enable us to work with the fields  $M$  or  $M'$  instead of  $K$ .

**Lemma 7.5.** Let  $K = \mathbb{Q}(\omega)$  and  $M = K(\sqrt[3]{\pi\theta})$ , where  $\pi, \theta \in O_K$  are prime, coprime to each other. Then  $\pi$  ramifies completely in  $M$ , and for  $\pi'$  over  $\pi$  in  $M$  we have

$$\left(\frac{\theta}{\pi}\right)_3 = \left(\frac{\theta}{\pi'}\right)_3,$$

meaning  $\theta^{\frac{N(\pi)-1}{3}} \equiv x \pmod{\pi}$  and  $\theta^{\frac{N(\pi')-1}{3}} \equiv x \pmod{\pi'}$  for the same  $x \in \{1, \omega, \omega^2\}$ , where  $N(\pi) = |O_K/\pi|$  and  $N(\pi') = |O_M/\pi'|$ . Furthermore, the same statement holds for  $M$  replaced by  $M'$ .

*Proof.* We will see in the proof of Lemma 7.4 that  $\pi$  ramifies completely. This implies that  $\pi'$  over  $\pi$  has inertial degree 1, and we have that the field extension induces an isomorphism  $O_K/\pi \rightarrow O_M/\pi'$ . Thus it holds that  $N(\pi) = N(\pi')$ . Now

$$\theta^{\frac{(N(\pi)-1)}{3}} \equiv x \pmod{\pi} \iff \theta^{\frac{(N(\pi)-1)}{3}} \equiv x \pmod{\pi'}$$

since  $\pi = O_K \cap \pi'$  and  $\theta \in O_K$ ,  $x \in \{1, \omega, \omega^2\}$ . The same argument will also work for  $M$  replaced by  $M'$ .  $\square$

Now we give the second proof of cubic reciprocity.

*Proof of cubic reciprocity.* We have shown that either  $M \subset L$  or  $M' \subset L$  is unramified. This allows us to look at  $(\frac{\pi}{\cdot}), (\frac{\theta}{\cdot})$  in the same field extension.

Assume  $M \subset L$  is unramified. We will see in the proof of Lemma 7.4 that  $\pi$  and  $\theta$  are both are totally ramified in  $M$ . This means we can write  $\pi = \pi'^3$  and  $\theta = \theta'^3$  for primes  $\pi', \theta'$  in  $M$ . Thus we obtain that  $\langle \sqrt[3]{\pi\theta} \rangle^3 = \langle \pi\theta \rangle = \langle \pi \rangle \cdot \langle \theta \rangle = \pi'^3 \cdot \theta'^3$ , so by uniqueness of factorisation we have  $\langle \sqrt[3]{\pi\theta} \rangle = \pi' \cdot \theta'$ . In particular  $\pi' \cdot \theta'$  is principal.

To simplify notation, let  $\chi(\alpha, \beta) := (\Phi_{L/M,1}(\alpha)(\sqrt[3]{\beta}))/\sqrt[3]{\beta} \in \{1, \omega, \omega^2\}$ , for  $\alpha \in I_M$ ,  $\beta \in \{\pi, \theta\}$ , where  $\Phi_{L/M,1}$  is the Artin map of the unramified field extension  $M \subset L$ . Notice that if  $\beta$  does not divide  $3\alpha$ , we have that  $\chi(\alpha, \beta) = (\frac{\beta}{\alpha})_3$  by equation (3). Also note that  $\chi(\alpha, \beta)$  is mutliplicative in the first variable since the Artin map is, and mutliplicative in the second variable by properties of field automorphisms. Furthermore, by construction, if  $\alpha \in P_M$  then  $\chi(\alpha, \beta) = 1$ .

Consider the following diagram, which is commutative by construction:

$$\begin{array}{ccc} & & \{1, \omega, \omega^2\} \\ & \nearrow \chi(\cdot, \pi) & \uparrow \sigma \mapsto \sigma(\sqrt[3]{\pi})/\sqrt[3]{\pi} \\ I_M/P_M & \xrightarrow{\Phi_{L/M,1}} & \text{Gal}(L/M) \\ & \searrow \chi(\cdot, \theta) & \downarrow \sigma \mapsto \sigma(\sqrt[3]{\theta})/\sqrt[3]{\theta} \\ & & \{1, \omega, \omega^2\} \end{array}$$

Notice that we have two ways of identifying  $\text{Gal}(L/M)$  with  $\{1, \omega, \omega^2\}$  by either using the image of  $\sqrt[3]{\pi}$  or  $\sqrt[3]{\theta}$ . But since  $\sqrt[3]{\pi\theta}$  is in  $M$  and thus fixed by every element of the Galois group, i.e.,  $\sigma(\sqrt[3]{\pi\theta})/\sqrt[3]{\pi\theta} = 1$  for all  $\sigma \in \text{Gal}(L/M)$ , we know that one will correspond to the inverse of the other:

$$\begin{array}{ccc} & & \{1, \omega, \omega^2\} \\ & \nearrow \chi(\cdot, \pi) & \uparrow \omega \mapsto \omega^{-1} \\ I_M/P_M & \searrow \chi(\cdot, \theta) & \\ & & \{1, \omega, \omega^2\} \end{array}$$

In particular, we get that

$$\chi(\cdot, \pi) = \chi(\cdot, \theta)^{-1}. \quad (5)$$

Moreover, since  $\pi' \cdot \theta' \in P_M$  we also obtain

$$\chi(\pi', \pi) \cdot \chi(\theta', \pi) = \chi(\pi' \theta', \pi) = 1. \quad (6)$$

Combining (5) and (6), we get

$$\chi(\pi', \theta)^{-1} \cdot \chi(\theta', \pi) = 1$$

which implies  $\chi(\theta', \pi) = \chi(\pi', \theta)$ . Recall that if  $\beta$  does not divide  $3\alpha$ , we have that  $\chi(\alpha, \beta) = \left(\frac{\beta}{\alpha}\right)_3$ . Hence we obtain

$$\left(\frac{\pi}{\theta'}\right)_3 = \left(\frac{\theta}{\pi'}\right)_3.$$

since  $\pi, \theta$  are coprime to each other and to 3 and we conclude by Lemma 7.5.

The same argument applies if  $M' \subset L$  is unramified. We can write  $\pi = \pi'^3$  and  $\theta = \theta'^3$  for primes  $\pi', \theta'$  in  $M'$ . Then  $\langle \sqrt[3]{\pi^2 \theta} \rangle^3 = \langle \pi^2 \theta \rangle = \langle \pi \rangle^2 \cdot \langle \theta \rangle = \pi'^6 \cdot \theta'^3$  so by uniqueness of factorisation we have that  $\pi'^2 \cdot \theta' = \langle \sqrt[3]{\pi \theta} \rangle$  is principal. Let  $\chi(\alpha, \beta)$  be as above, but with the Artin map  $\Phi_{L/M', 1}$  of  $M' \subset L$ . The corresponding diagram is:

$$\begin{array}{ccc} & & \{1, \omega, \omega^2\} \\ & \nearrow \chi(\cdot, \pi^2) & \uparrow \sigma \mapsto \sigma(\sqrt[3]{\pi^2}) / \sqrt[3]{\pi^2} \\ I_{M'}/P_{M'} & \xrightarrow{\Phi_{L/M', 1}} & \text{Gal}(L/M') \\ & \searrow \chi(\cdot, \theta) & \downarrow \sigma \mapsto \sigma(\sqrt[3]{\theta}) / \sqrt[3]{\theta} \\ & & \{1, \omega, \omega^2\} \end{array}$$

Just like before  $\text{Gal}(L/M')$  fixes  $\sqrt[3]{\pi^2 \theta}$ , so we get  $\chi(\cdot, \pi^2) = \chi(\cdot, \theta)_3^{-1}$ . Since  $\pi'^2 \cdot \theta' \in P_M$  we again have that  $\chi(\pi', \pi^2)^2 \cdot \chi(\theta', \pi^2) = \chi(\pi'^2 \theta', \pi^2) = 1$ .

Combining these we obtain  $\chi(\pi', \theta)^{-2} \cdot \chi(\theta', \pi^2) = 1$  so  $\chi(\theta', \pi^2) = \chi(\pi', \theta)^2$ . Since both sides are in  $\{1, \omega, \omega^2\}$  we deduce that  $\chi(\theta', \pi) = \chi(\pi', \theta)$ . Finally, using that  $\pi, \theta$  are coprime to each other and to 3 this implies that  $\left(\frac{\pi}{\theta'}\right)_3 = \left(\frac{\theta}{\pi'}\right)_3$  and we conclude by Lemma 7.5.  $\square$

Let us now prove Lemma 7.4.

**Remark 7.6.** In order to simplify the calculations we will use Remark 3.8 in the following way. Let  $\mathfrak{p}$  be some prime in  $K$  and let  $I = \{K(\sqrt[3]{\pi}), K(\sqrt[3]{\theta}), M, M'\}$ . If there exists two different intermediate fields  $A, B \in I$  such that  $\mathfrak{p}$  is ramified in  $A$  and unramified in  $B$ , then we know that for all  $C \in I \setminus \{B\}$

$$\begin{aligned} & \mathfrak{p} \text{ ramifies in } K \subset C \\ & \text{primes over } \mathfrak{p} \text{ are unramified in } C \subset L \end{aligned}$$

Note that a prime is totally ramified if it is ramified in Galois extensions of degree 3. This statement is then a consequence of the existence a unique largest intermediate field where  $\mathfrak{p}$  is unramified, which is true in all abelian extensions as seen in Lemma 3.8.

*Proof of Lemma 7.4.* We start by showing that both  $M \subset L$  and  $M' \subset L$  are unramified at primes over  $\pi$  and  $\theta$ . This follows by Remark 7.6 since

- (i)  $\theta$  is unramified in  $K(\sqrt[3]{\pi})$  since  $\theta$  is coprime to  $\pi$  and 3
- (ii)  $\theta$  is totally ramified in  $K(\sqrt[3]{\theta})$  since  $\langle \theta \rangle = \langle \sqrt[3]{\theta} \rangle^3$
- (iii)  $\pi$  is unramified in  $K(\sqrt[3]{\theta})$  since  $\pi$  is coprime to  $\theta$  and 3
- (iv)  $\pi$  is totally ramified in  $K(\sqrt[3]{\pi})$  since  $\langle \pi \rangle = \langle \sqrt[3]{\pi} \rangle^3$ .

Note that this also implies that  $\pi, \theta$  are ramified and thus totally ramified (we have a Galois extension of degree 3) in  $K \subset M$  and  $K \subset M'$  by Remark 7.6.

We now turn to the primes over 3 or rather  $1 - \omega$ . This will be a bit more complicated and we quote the following result from Milne's lecture notes on class field theory ([Mil20]) without proof.

**Lemma 7.7** ([Mil20], Remark A.6). Let  $p$  be a rational prime and  $K$  a number field containing a primitive  $p$ -th root of unity  $\zeta_p$ . Furthermore, let  $\alpha \in K^*$  be coprime to  $p$  and not a third power in  $K$ .

$K \subset K(\sqrt[p]{\alpha})$  is unramified at all primes over  $p \iff X^3 \equiv \alpha \pmod{p(1 - \zeta_p)}$  has a solution in  $O_K$ .

For our case  $p = 3$  let  $\lambda = 1 - \omega$  and recall that our primes  $\pi, \theta$  are  $1 \pmod{3}$ . We write

$$\begin{aligned}\pi &= 1 + a\lambda^2 \\ \theta &= 1 + b\lambda^2\end{aligned}$$

for  $a, b \in O_K$ .

Notice that  $|O_K/\langle \lambda \rangle| = 3$  so there are each three possibilities for  $a, b$  modulo  $\lambda$ . We can distinguish between three different cases:

- (i)  $a \equiv 0 \pmod{\lambda}$  or  $b \equiv 0 \pmod{\lambda}$

This implies  $\pi \equiv 1 \pmod{\lambda^3}$  or  $\theta \equiv 1 \pmod{\lambda^3}$ . We have  $L = M(\sqrt[3]{\pi})$  and also  $L = M(\sqrt[3]{\theta})$ , thus by Lemma 7.7,  $M \subset L$  is unramified at primes over 3. In fact, by the same argument,  $M' \subset L$  is unramified at primes over 3.

- (ii)  $a, b \not\equiv 0 \pmod{\lambda}$  and  $a \equiv b \pmod{\lambda}$

Then we get  $\pi^2\theta \equiv 1 + (2a + b)\lambda^2 \pmod{\lambda^4}$ ,  $2a + b \equiv 3a \equiv 0 \pmod{\lambda}$  and thus

$$\pi^2\theta \equiv 1 \pmod{\lambda^3}.$$

Since  $L = M(\sqrt[3]{\pi^2\theta})$ , by Lemma 7.7,  $M \subset L$  is unramified at primes over 3.

(iii)  $a, b \not\equiv 0 \pmod{\lambda}$  and  $a \not\equiv b \pmod{\lambda}$ , i.e.  $a + b \equiv 0 \pmod{\lambda}$

Then we get  $\pi\theta \equiv 1 + (a + b)\lambda^2 \pmod{\lambda^4}$ ,  $a + b \equiv 0 \pmod{\lambda}$  and thus

$$\pi\theta \equiv 1 \pmod{\lambda^3}.$$

Since  $L = M'(\sqrt[3]{\pi\theta})$ , by Lemma 7.7,  $M' \subset L$  is unramified at primes over 3.

In cases (i) both  $M \subset L$  and  $M' \subset L$  are unramified, whereas in case (ii)  $M \subset L$  is unramified and in (iii)  $M' \subset L$  is unramified. □

## Appendix A: Local Class Field Theory

We will give a very short overview of some concepts and theorems of local class field theory. We will assume that the reader is familiar with the group cohomology of Tate cohomology groups, for a detailed introduction to group cohomology we refer to Milne ([Mil20], Chapter II). We first start with the existence of the local Artin map. Some notation:

- (i) In this section, if not further specified, we assume  $K$  to be a local field. We generally denote its maximal ideal by  $\mathfrak{p}$  and by  $\pi$  some prime element. We call  $u \in K$  a unit in  $K$  if  $u$  is a unit in the valuation ring, i.e., the ring of integers  $O_K$ .
- (ii) We denote by  $K^{\text{ab}}$  the union of all finite abelian extensions of  $K$ . Its Galois group is the quotient of the Galois group of a separable algebraic closure of  $K$  by its commutator subgroup.
- (iii) For a finite unramified extension  $L$  of  $K$  we have that  $\text{Gal}(L/K) \cong \text{Gal}(l/k)$ , where  $l, k$  are the corresponding residue fields. We then have a unique generator of  $\text{Gal}(L/K)$  that maps to the Frobenius element in  $\text{Gal}(l/k)$  which we denote by  $\text{Frob}_{L/K} \in \text{Gal}(L/K)$  (and also call Frobenius element). Also note that any finite unramified extension of local fields is automatically abelian.
- (iv) For a finite abelian extension we denote the norm map by

$$N_{L/K} : L \longrightarrow K$$

$$\alpha \mapsto \prod_{\sigma \in \text{Gal}(L/K)} \sigma \alpha.$$

- (v) We denote by  $H_T^r(G, M)$  the  $r$ -th Tate cohomology group and by  $H^r(G, M)$  the  $r$ -th cohomology group of a  $G$ -module  $M$  (an abelian group with a  $G$ -action). In this section,  $G$  will always be a Galois group, and the action will either be the obvious action of the Galois group or the trivial action (if the  $G$ -module is  $\mathbb{Z}$ ).

We first state the existence of the local Artin map. Similar to the global Artin map that we saw earlier, it is a homomorphism that acts like the Frobenius element on unramified field extension.

**Theorem 7.8** (The local Artin map, [Mil20], Theorem 3.4). For every local field  $K$ , there exists a homomorphism

$$\phi_K : K^* \longrightarrow \text{Gal}(K^{\text{ab}}/K),$$

with the following properties:

- (i) for every prime element  $\pi$  of  $K$  and every finite unramified extension  $L$  of  $K$ ,  $\phi_K(\pi)|_L = \text{Frob}_{L/K}$
- (ii) for every finite abelian extension  $L$  of  $K$ ,  $N_{L/K}L^*$  is contained in the kernel of  $a \mapsto \phi_K(a)|_L$  and  $\phi_K$  induces an isomorphism

$$\phi_{L/K} : K^*/N_{L/K}L^* \xrightarrow{\sim} \text{Gal}(L/K).$$

**Remark 7.9.** The local Artin map can be constructed by piecing together to local Artin maps of finite abelian extensions  $L$  of  $K$ . For a finite abelian extensions  $L$  of  $K$ , we get the local Artin map  $\phi_{L/K}$  as follows: Tate's theorem (see Milne ([Mil20], Chapter III)) gives us an isomorphism

$$H_T^r(\mathrm{Gal}(L/K), \mathbb{Z}) \xrightarrow{\sim} H_T^{r+2}(\mathrm{Gal}(L/K), L^*)$$

$$x \mapsto x \cup u_{L/K}$$

induced by the cup-product with  $u_{L/K}$ , where  $u_{L/K}$  is the *fundamental class* of  $L/K$  (see Milne ([Mil20], p.111)), a generator of  $H^2(\mathrm{Gal}(L/K), L^*)$ . In the case  $r = -2$  we get,

$$\begin{array}{ccc} H_T^{-2}(\mathrm{Gal}(L/K), \mathbb{Z}) & \xrightarrow{\sim} & H_T^0(\mathrm{Gal}(L/K), L^*) \\ \wr & & \wr \\ \mathrm{Gal}(L/K) & & K^*/N_{L/K}L^* \end{array}$$

The left isomorphism will follow from the long exact sequence of cohomology groups of a short exact sequence of  $G$ -modules (see Milne([Mil20], p.77)), the right one follows from definition of Tate cohomology groups. The inverse to this map is the local Artin map  $\phi_{L/K}$ .

**Proposition 7.10** ([Mil20], Lemma 3.7). If  $L/K$  is finite and unramified, then  $\phi_{L/K}$  sends  $a \in K^*$  to  $\mathrm{Frob}^{\mathrm{ord}(a)}$ , where  $\mathrm{Frob}$  is the Frobenius element in  $\mathrm{Gal}(L/K)$  and  $\mathrm{ord}(a)$  such that  $a = u\pi^{\mathrm{ord}(a)}$  for  $u$  a unit in  $K$  and  $\pi$  a prime element in  $K$ .

*Proof.* See Milne ([Mil20], p.109). □

For our discussion of the power residue symbols we will unfortunately need one statement of the idèle-theoretic formulation of global class field theory. The reader who is unfamiliar with idèles can simply think of the global idèle-theoretic Artin map as pieced together by the local Artin maps.

**Definition 7.11** (Adèles and Idèles). Let  $K$  be a number field. An adèle of  $K$  is a family

$$\alpha = (\alpha_{\mathfrak{p}})$$

of elements  $\alpha_{\mathfrak{p}} \in K_{\mathfrak{p}}$  where  $\mathfrak{p}$  runs through all finite and infinite primes, such that  $\alpha_{\mathfrak{p}}$  is integral in  $K_{\mathfrak{p}}$ , i.e., in the valuation ring  $O_{\mathfrak{p}}$ , for almost all  $\mathfrak{p}$ . The adèles form a ring  $\mathbb{A}_K$  with componentwise addition and multiplication.

The group of idèle of  $K$  is the unit group  $\mathbb{A}_K^*$ . An idèle is a family

$$\alpha = (\alpha_{\mathfrak{p}})$$

of elements  $\alpha_{\mathfrak{p}} \in K_{\mathfrak{p}}^*$  such that  $\alpha_{\mathfrak{p}}$  is a unit in the ring of integers in  $K_{\mathfrak{p}}$  for almost all  $\mathfrak{p}$ . We call an idèle *principal* if it is of the form  $(\alpha, \alpha, \alpha, \dots)$  for some  $\alpha \in K^*$ .

We will now assume the existence of the global Artin map defined on the idèle group of  $K$ . For a proof we refer to Milne ([Mil20], Chapter V). Again, it will be enough to just think of the global Artin map as pieced together by the local Artin maps at every prime of  $K$ . The next theorem captures this idea.

**Theorem 7.12** ([Neu13], Chapter V Cor.5.7). Let  $K \subset L$  be an abelian extension of number fields and  $\alpha = (\alpha_{\mathfrak{p}})$  an idèle of  $K$ , then

$$\Phi_{L/K}(\alpha) = \prod_{\mathfrak{p}} \phi_{L_{\mathfrak{p}}/K_{\mathfrak{p}}}(\alpha_{\mathfrak{p}}),$$

where  $\Phi_{L/K}$  is the global idèle-theoretic Artin map. In particular, for a principal idèle  $(\alpha, \alpha, \alpha, \dots)$  for  $\alpha \in K^*$  we have

$$\Phi_{L/K}(\alpha, \dots) = \prod_{\mathfrak{p}} \phi_{L_{\mathfrak{p}}/K_{\mathfrak{p}}}(\alpha) = 1.$$

*Proof.* See Neukirch ([Neu13], p.393). □

## Appendix B: The Hilbert Symbol

In this section we define the Hilbert symbol and prove some of its properties, mostly following Neukirch ([Neu13], Chapter V §3). Throughout this section let  $K$  be a local field with maximal ideal  $\mathfrak{p}$  containing a primitive  $n$ -th root of unity and let  $\mu_n$  denote the group of  $n$ -th roots of unity.

Let  $L = K(\sqrt[n]{K^*})$  be the maximal abelian extension with Galois group of exponent  $n$ . In order to define the Hilbert symbol we use the following two isomorphisms:

(i) The local Artin map

$$\begin{aligned} K^*/K^{*n} &\longrightarrow \text{Gal}(L/K), \\ a &\mapsto \phi_{L/K}(a) \end{aligned} \tag{7}$$

where we used that  $N_{L/K}(L^*) = K^{*n}$  (see Neukirch ([Neu13], p.322)).

(ii)

$$\begin{aligned} K^*/K^{*n} &\longrightarrow \text{Hom}(\text{Gal}(L/K), \mu_n) \\ b &\mapsto \chi_b : \tau \mapsto \tau(\sqrt[n]{b})/\sqrt[n]{b}. \end{aligned} \tag{8}$$

We obtain this isomorphism by considering the long exact sequence of cohomology of the short exact sequence

$$0 \longrightarrow \mu_n \longrightarrow L^* \xrightarrow{x \mapsto x^n} L^* \longrightarrow 0$$

In particular we get

$$\dots \longrightarrow H^0(\text{Gal}(L/K), L^*) \xrightarrow{x \mapsto x^n} H^0(\text{Gal}(L/K), L^*) \longrightarrow H^1(\text{Gal}(L/K), \mu_n) \longrightarrow 0 \longrightarrow \dots$$

where we used Hilbert 90, namely that  $H^1(\text{Gal}(L/K), L^*) = 0$ . Then we can use that  $H^0(\text{Gal}(L/K), L^*) = K^*$  and  $H^1(\text{Gal}(L/K), \mu_n) = \text{Hom}(\text{Gal}(L/K), \mu_n)$ , since the action of  $\text{Gal}(L/K)$  on  $\mu_n$  is trivial. For more details see Milne ([Mil20], Chapter II).

**Definition 7.13** (Hilbert Symbol). Let  $K$  be a local field containing a primitive  $n$ -th root of unity. Let  $L = K(\sqrt[n]{K^*})$  be the maximal abelian extension of exponent  $n$ . We have a pairing

$$\begin{aligned} \text{Gal}(L/K) \times \text{Hom}(\text{Gal}(L/K), \mu_n) &\longrightarrow \mu_n \\ (\sigma, \chi) &\mapsto \chi(\sigma) \end{aligned}$$

Using (7) and (8), we obtain a pairing

$$\left(\frac{\cdot, \cdot}{\mathfrak{p}}\right)_n : K^*/K^{*n} \times K^*/K^{*n} \longrightarrow \mu_n$$

which we will call the  $n$ -th Hilbert symbol.

**Remark 7.14.** The notation using the maximal ideal  $\mathfrak{p}$  makes more sense when we start regarding the completions of number fields at different places  $\mathfrak{p}$ . If  $K$  is a number field we will mostly use  $\left(\frac{\cdot, \cdot}{\mathfrak{p}}\right)_n$  to denote the Hilbert symbol defined on the completion of  $K$  at the place  $\mathfrak{p}$ , for  $\mathfrak{p}$  a finite or infinite prime of  $K$ .

**Remark 7.15.** The Hilbert symbol also arises naturally as a cup-product of cohomology groups

$$\begin{array}{ccccc} H^1(\text{Gal}(L/K), \mu_n) & \times & H^1(\text{Gal}(L/K), \mu_n) & \longrightarrow & H^2(\text{Gal}(L/K), \mu_n \otimes \mu_n) \\ \wr \downarrow & & \wr \downarrow & & \wr \downarrow \\ K^*/K^{*n} & & K^*/K^{*n} & & \mu_n \end{array}$$

For details see Milne ([Mil20], p.113).

**Proposition 7.16** ([Neu13], Chapter V Prop.3.1). Let  $a, b \in K^*$ , then the Hilbert symbol is given by

$$\phi_{L/K}(a)(\sqrt[n]{b}) = \left(\frac{a, b}{\mathfrak{p}}\right)_n \sqrt[n]{b}.$$

*Proof.* We simply use the definition of the Hilbert symbol. Let  $\chi_b \in \text{Hom}(\text{Gal}(L/K), \mu_n)$  be given by  $\chi_b(\tau) = \tau(\sqrt[n]{b})/\sqrt[n]{b}$ . We then have

$$\left(\frac{a, b}{\mathfrak{p}}\right)_n \sqrt[n]{b} = \chi_b(\phi_{L/K}(a)) \sqrt[n]{b} = \frac{\phi_{L/K}(a)(\sqrt[n]{b})}{\sqrt[n]{b}} \sqrt[n]{b} = \phi_{L/K}(a)(\sqrt[n]{b}) \quad \square$$

**Remark 7.17.** We use  $L = K(\sqrt[n]{K^*})$  here, but since any field extension with Galois group of exponent  $n$  is contained in  $L$ , the statements hold for any field extension with Galois group of exponent  $n$ .

The Hilbert symbol satisfies the following properties.

**Proposition 7.18** ([Neu13], Chapter VI Theorem 8.1). Let  $\alpha, \alpha', \beta, \beta' \in K^*$ .

- (i) Bi-multiplicativity:  $\left(\frac{\alpha\alpha', \beta}{\mathfrak{p}}\right)_n = \left(\frac{\alpha, \beta}{\mathfrak{p}}\right)_n \cdot \left(\frac{\alpha', \beta}{\mathfrak{p}}\right)_n$  and  $\left(\frac{\alpha, \beta\beta'}{\mathfrak{p}}\right)_n = \left(\frac{\alpha, \beta}{\mathfrak{p}}\right)_n \cdot \left(\frac{\alpha, \beta'}{\mathfrak{p}}\right)_n$
- (ii) Skew-symmetry:  $\left(\frac{\alpha, \beta}{\mathfrak{p}}\right)_n = \left(\frac{\beta, \alpha}{\mathfrak{p}}\right)_n^{-1}$

- (iii)  $\left(\frac{\alpha, -\alpha}{\mathfrak{p}}\right)_n = 1$  and  $\left(\frac{\alpha, 1-\alpha}{\mathfrak{p}}\right)_n = 1$
- (iv)  $\left(\frac{\alpha, \beta}{\mathfrak{p}}\right)_n = 1 \iff \alpha$  is a norm of the extension  $K \subset K(\sqrt[n]{\beta})$ , i.e.,  $\alpha \in N_{K(\sqrt[n]{\beta})/K} K(\sqrt[n]{\beta})^*$
- (v) Non-degeneracy:  $\left(\frac{\alpha, \beta}{\mathfrak{p}}\right)_n = 1 \ \forall \beta \in K^* \iff a \in K^{*n}$

*Proof.* See Neukirch ([Neu13], p.334).

Note that (iv) follows directly using Proposition 7.16 since

$$\phi_{K(\sqrt[n]{\beta})/K}(a)(\sqrt[n]{\beta}) = \sqrt[n]{\beta}$$

implies that  $\phi_{K(\sqrt[n]{\beta})/K}(a)$  is the identity. But the kernel of the local Artin map consists of the norms of the extension, so we deduce (iii).  $\square$

**Proposition 7.19** (Product formula). Let  $K$  be a number field containing the  $n$ -th roots of unity. For  $\alpha, \beta \in K^*$  we have

$$\prod_{\mathfrak{p}} \left(\frac{\alpha, \beta}{\mathfrak{p}}\right)_n = 1,$$

where  $\mathfrak{p}$  runs over all finite and infinite primes.

*Proof.* Using Lemma 7.12 we obtain

$$\prod_{\mathfrak{p}} \left(\frac{\alpha, \beta}{\mathfrak{p}}\right)_n \sqrt[n]{\beta} = \left[ \prod_{\mathfrak{p}} \phi_{K_{\mathfrak{p}}(\sqrt[n]{\beta})/K_{\mathfrak{p}}}(\alpha) \right] (\sqrt[n]{\beta}) = \Phi_{K(\sqrt[n]{\beta})/K}(\alpha)(\sqrt[n]{\beta}) = \sqrt[n]{\beta}$$

The last step follows since we have  $\Phi_{K(\sqrt[n]{\beta})/K}(\alpha) = 1$  for  $\alpha \in K^*$  by Theorem 7.12, where we consider  $\alpha$  to be the idèle  $(\alpha, \alpha, \dots)$ .  $\square$

**Example 7.20.** The Hilbert symbol has a very nice description in the case  $K = \mathbb{Q}_p$  and  $n = 2$ . For  $a, b \in \mathbb{Q}_p^*$  we have

$$\left(\frac{a, b}{p}\right)_2 = \begin{cases} +1 & \text{if } z^2 = ax^2 + by^2 \text{ has a nontrivial solution in } \mathbb{Q}_p \\ -1 & \text{otherwise} \end{cases}$$

This is seen very easily using Proposition 7.2, namely

$$\begin{aligned} \left(\frac{a, b}{p}\right)_2 = +1 &\iff a \text{ is a norm of the extension } \mathbb{Q}_p \subset \mathbb{Q}_p(\sqrt{b}) \\ &\iff a = N(z + y\sqrt{b}) = (z + y\sqrt{b})(z - y\sqrt{b}) = z^2 - by^2 \text{ has a solution in } \mathbb{Q}_p \\ &\iff z^2 = ax^2 + by^2 \text{ has a nontrivial solution in } \mathbb{Q}_p \end{aligned}$$

**Proposition 7.21.** Let  $K$  be number field containing a  $n$ -th root of unity,  $\mathfrak{p}$  be a prime in  $K$  coprime to  $n$  and  $\alpha \in K$  coprime to  $\mathfrak{p}$  and  $n$ . Let  $K_{\mathfrak{p}}$  denote the completion of  $K$  at  $\mathfrak{p}$ . Furthermore, let  $u$  be a unit in  $K_{\mathfrak{p}}$ . Then

$$\left(\frac{u, \alpha}{\mathfrak{p}}\right)_n = 1,$$

where the Hilbert symbol is defined on  $K_{\mathfrak{p}}^*/K_{\mathfrak{p}}^{*n} \times K_{\mathfrak{p}}^*/K_{\mathfrak{p}}^{*n}$ .

*Proof.* We have that  $K_{\mathfrak{p}} \subset K_{\mathfrak{p}}(\sqrt[n]{\alpha})$  is unramified, since  $K \subset K(\sqrt[n]{\alpha})$  is unramified at  $\mathfrak{p}$ . Indeed,  $K \subset K(\sqrt[n]{\alpha})$  is only possibly ramified at primes dividing  $n$  or  $\alpha$ . We can now use that for an unramified field extension  $K \subset L$  of local fields,  $N_{L/K}(U_L) = U_K$ , where  $U_L, U_K$  are the group of units in  $K, L$ , for details we refer to Milne ([Mil20], p.98). Thus every unit in  $K_{\mathfrak{p}}$  is a norm in the extension  $K_{\mathfrak{p}} \subset K_{\mathfrak{p}}(\sqrt[n]{\alpha})$  and by Proposition 7.18 part (iv) we can deduce the statement.  $\square$

## Appendix C: The Legendre Symbol (revisited)

Let us now revisit the power residue symbol. Let  $K$  be a number field containing an  $n$ -th root of unity. Let  $\mathfrak{p}$  be a prime in  $K$ , coprime to  $n$  and let  $\alpha \in K$  coprime to  $\mathfrak{p}$ . Recall that the  $n$ -th power Legendre symbol is defined to be the  $n$ -th root of unity satisfying

$$\alpha^{(N(\mathfrak{p})-1)/n} \equiv \left(\frac{\alpha}{\mathfrak{p}}\right)_n \pmod{\mathfrak{p}}.$$

Using the Hilbert symbol we get a new description of the  $n$ -th power Legendre symbol.

**Proposition 7.22** ([Neu13], Chapter V Prop.3.1). Let  $\alpha, \mathfrak{p}$  be as above and let  $K_{\mathfrak{p}}$  denote the completion of  $K$  at  $\mathfrak{p}$ . Furthermore, let  $\pi$  be a prime element of  $K_{\mathfrak{p}}$ . Then

$$\left(\frac{\alpha}{\mathfrak{p}}\right)_n = \left(\frac{\pi, \alpha}{\mathfrak{p}}\right)_n,$$

where the Hilbert symbol is defined on  $K_{\mathfrak{p}}^*/K_{\mathfrak{p}}^{*n} \times K_{\mathfrak{p}}^*/K_{\mathfrak{p}}^{*n}$ .

*Proof.* We have shown in Lemma 7.16 that

$$\left(\frac{\pi, \alpha}{\mathfrak{p}}\right)_n \sqrt[n]{\alpha} = \phi_{K_{\mathfrak{p}}(\sqrt[n]{\alpha})/K_{\mathfrak{p}}}(\pi)(\sqrt[n]{\alpha}).$$

By Lemma 7.10 the local Artin map maps a prime element to the Frobenius element in  $\text{Gal}(K_{\mathfrak{p}}(\sqrt[n]{\alpha})/K_{\mathfrak{p}})$  (notice that this field extension is unramified since  $n$  and  $\alpha$  are coprime to  $\mathfrak{p}$ ). Thus

$$\left(\frac{\pi, \alpha}{\mathfrak{p}}\right)_n \sqrt[n]{\alpha} = \text{Frob}_{K_{\mathfrak{p}}(\sqrt[n]{\alpha})/K_{\mathfrak{p}}}(\sqrt[n]{\alpha}) \equiv \sqrt[n]{\alpha}^{N(\mathfrak{p})} \equiv (\alpha^{1/n})^{N(\mathfrak{p})-1} \sqrt[n]{\alpha} \pmod{\mathfrak{p}}$$

and we deduce that

$$\left(\frac{\pi, \alpha}{\mathfrak{p}}\right)_n \equiv (\alpha^{1/n})^{N(\mathfrak{p})-1} \pmod{\mathfrak{p}}$$

which coincides with the definition of the  $n$ -th power Legendre symbol.  $\square$

Also remember that we can extend the Legendre symbol by multiplicativity. For  $\mathfrak{m}$  a modulus in  $K$  such that  $n\alpha$  divides  $\mathfrak{m}$ , we have the homomorphism

$$\left(\frac{\alpha}{\cdot}\right)_n : I_K(\mathfrak{m}) \longrightarrow \mu_n$$

$$\left(\frac{\alpha}{\mathfrak{a}}\right)_n = \prod_{i=1}^r \left(\frac{\alpha}{\mathfrak{p}_i}\right)_n^{\nu_i}$$

for  $\mathfrak{a} = \mathfrak{p}_1^{\nu_1} \dots \mathfrak{p}_r^{\nu_r}$ .

Now we can prove the general reciprocity law, which we previously called the strong reciprocity law.

**Theorem 7.23** ([Neu13], Chapter VI Theorem 8.3). Let  $K$  be a number field containing a primitive  $n$ -th root of unity  $\zeta_n^k$  and let  $\alpha, \beta \in O_K$  be coprime to each other and to  $n$ . We will write  $\mathfrak{p} \mid n$  if  $\mathfrak{p}$  appears in the prime decomposition of  $\langle n \rangle$  in  $K$ . Then

$$\left(\frac{\alpha}{\beta}\right)_n \cdot \left(\frac{\beta}{\alpha}\right)_n^{-1} = \prod_{\mathfrak{p} \mid n\infty} \left(\frac{\alpha, \beta}{\mathfrak{p}}\right)_n,$$

where  $\infty$  is the product of the real infinite primes of  $K$ .

*Proof.* If  $\mathfrak{p}$  appears in the prime decomposition of  $\langle \beta \rangle$  with exponent  $\nu_{\mathfrak{p}}(\beta)$  then  $\mathfrak{p} \nmid \alpha n\infty$ . Now if  $\pi$  is a prime element in  $K_{\mathfrak{p}}$  then we have  $\beta = u\pi^{\nu_{\mathfrak{p}}(\beta)}$  for  $u$  a unit in  $K_{\mathfrak{p}}$ . Therefore

$$\left(\frac{\alpha}{\mathfrak{p}}\right)_n^{\nu_{\mathfrak{p}}(\beta)} = \left(\frac{\pi, \alpha}{\mathfrak{p}}\right)_n^{\nu_{\mathfrak{p}}(\beta)} = \left(\frac{\pi^{\nu_{\mathfrak{p}}(\beta)}, \alpha}{\mathfrak{p}}\right)_n = \left(\frac{\beta, \alpha}{\mathfrak{p}}\right)_n$$

where we used that  $\left(\frac{u, \alpha}{\mathfrak{p}}\right)_n = 1$  by Proposition 7.21.

Similarly, if  $\mathfrak{p}$  appears in the prime decomposition of  $\langle \alpha \rangle$  with exponent  $\nu_{\mathfrak{p}}(\alpha)$  then  $\mathfrak{p} \nmid \beta n\infty$  and

$$\left(\frac{\beta}{\mathfrak{p}}\right)_n^{\nu_{\mathfrak{p}}(\alpha)} = \left(\frac{\alpha, \beta}{\mathfrak{p}}\right)_n.$$

Furthermore, we have that if  $\mathfrak{p} \nmid \alpha\beta n\infty$  then by using Proposition 7.21 again we get

$$\left(\frac{\alpha, \beta}{\mathfrak{p}}\right)_n = 1 \tag{9}$$

since  $\alpha, \beta$  are units in  $K_{\mathfrak{p}}$ . If we write  $\mathfrak{p} \mid \beta$  for  $\mathfrak{p}$  appearing in the prime decomposition of  $\beta$  with exponent  $\nu_{\mathfrak{p}}(\beta)$  and the same for  $\alpha$ , we get

$$\begin{aligned} \left(\frac{\alpha}{\beta}\right)_n \cdot \left(\frac{\beta}{\alpha}\right)_n^{-1} &= \prod_{\mathfrak{p} \mid \beta} \left(\frac{\alpha}{\mathfrak{p}}\right)_n^{\nu_{\mathfrak{p}}(\beta)} \cdot \prod_{\mathfrak{p} \mid \alpha} \left(\frac{\beta}{\mathfrak{p}}\right)_n^{-\nu_{\mathfrak{p}}(\alpha)} = \prod_{\mathfrak{p} \mid \beta} \left(\frac{\beta, \alpha}{\mathfrak{p}}\right)_n \cdot \prod_{\mathfrak{p} \mid \alpha} \left(\frac{\alpha, \beta}{\mathfrak{p}}\right)_n^{-1} \\ &= \prod_{\mathfrak{p} \mid \alpha\beta} \left(\frac{\beta, \alpha}{\mathfrak{p}}\right)_n \stackrel{(9)}{=} \prod_{\substack{\mathfrak{p} \mid \alpha\beta \\ \mathfrak{p} \nmid n\infty}} \left(\frac{\beta, \alpha}{\mathfrak{p}}\right)_n \cdot \prod_{\mathfrak{p} \nmid \alpha\beta n\infty} \left(\frac{\beta, \alpha}{\mathfrak{p}}\right)_n = \prod_{\mathfrak{p} \nmid n\infty} \left(\frac{\beta, \alpha}{\mathfrak{p}}\right)_n \end{aligned}$$

Using the product formula from Lemma 7.19

$$\prod_{\mathfrak{p}} \left(\frac{\alpha, \beta}{\mathfrak{p}}\right)_n = 1,$$

we obtain

$$\left(\frac{\alpha}{\beta}\right)_n \cdot \left(\frac{\beta}{\alpha}\right)_n^{-1} = \prod_{\mathfrak{p} \nmid n\infty} \left(\frac{\beta, \alpha}{\mathfrak{p}}\right)_n = \prod_{\mathfrak{p} | n\infty} \left(\frac{\beta, \alpha}{\mathfrak{p}}\right)_n^{-1} = \prod_{\mathfrak{p} | n\infty} \left(\frac{\alpha, \beta}{\mathfrak{p}}\right)_n$$

□

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